Uncertainty Quantification of a Compartmental Human Cardiovascular Model

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Declaration

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Abstract

Uncertainty quantification deals with the uncertainty of input parameters in mathematical models and simulations. Sensitivity analysis is one of many ways of doing uncertainty quantification but will be the focus in this project. Mechanistic models are used to represent the cardiovascular system, which can be accurate when supplied with input parameters from clinical data. Lumped parameter models are a methodology of representing cardiovascular systems and can be personalised for forming clinical decisions based on the patient's input parameters via sensitivity analysis. However, the computational time for models simulating high number of samples can be expensive, thus an implementation of a surrogate model is utilised. The workflow presented in this dissertation explores the possible methods one can use based on the review of multiple literature in this field.

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Thank you to my supervisor, Xu Xu, who helped me every step of the way. I hope I made you proud. I am wishing you and your family all the best, I hope they all do well!

To my Appa, Amma and Shagilan and Family, I just simply want to say thank you, you have provided all the support and love you could possibly give me. I love you so much and cannot wait to help you like you helped me.

To the friends that will leave, I wish you you guys all the best and wish to stay in contact with all of you guys no matter where you are in this world. Hope your dreams come true and hope you guys have a happy life

To the friends that will stay, I hope we can spend more time making memories and continue having fun until it is our time to depart.

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Chapter 1

Introduction

1.1 Overview of the Project

Cardiovascular disease is a significant cause of mortality and morbidity, accounting for a quarter of all deaths in the UK. It is the leading global cause of death, with an estimated 18 million deaths each year (over 30% of all global deaths) [4].

In the current landscape of in-silico physiology models in the clinic, there is a personalization problem, where models need to be adapted for individual patients, and an inverse problem, where we need to find the correct input parameters for each patient. The models tend to be computationally intensive as they use heavy calculations, which can be an issue where, in a healthcare setting, resources and time are limited.

The aim is to use sensitivity analysis to view how the uncertainty in a model input attributes to the uncertainty in the model's output and to see which input variable contributes the most to the output's behaviour and which are non-influential.

Surrogate modelling will be used to reduce computational cost, which will be an approximation of the original computational model. The surrogate model is built to mimic the original model but with less computational cost, which is constructed based on a limited number of runs of the true model. In order to perform model personalisation, uncertainty quantification analysis of the model needs to be done.

Sensitivity analysis is required to determine the effects of input parameters upon specific model outputs. Then, surrogate modelling is required to reduce the computational cost, using techniques such as polynomial chaos expansion.

1.2 Aims and Objectives

This project aims to form a model that is able to compute input parameters that gives detail to the cardiovascular model in the body, and then reduce the complexity of the model to improve its run time. In clinical settings, the model should aim to be accurate and efficient to run as results are needed within a short time period. The project's objectives are:

- Form a mechanistic model, using coupled ordinary differential equations that are in need to be solved.
- Perform sensitivity analysis on the model.
- Create a surrogate model of the current model.
- Increase the complexity over time. From one chamber to two chambers to four chambers.
- Create a surrogate model of the current model again.
- Then perform sensitivity analysis of that model.

Chapter 2

Literature Survey

2.1 Background

Mechanistic models enable the study of regulatory mechanisms implicated in various biological processes, like the cardiovascular model [5]. They provide a way to analyse the dynamics of the systems they describe and provide insight into the emerging behaviour of the system. One of the mathematical frameworks to be used from mechanistic models is called ordinary differential equations, which are useful to generate quantitative predictions. ODE-based modelling requires a steep learning curve as it relies on complex mathematical equations. The lumped parameter model is a form of mechanistic model. It is simple; however, it does not tend to give accurate results.

The process begins with a Zero-D model, which is formulated as a coupled system of ordinary differential equations that can be solved using numerical approximation methods. It is possible to use 1-D, 2-D or 3-D models, however there has to be a compromise in the accuracy and efficiency needed to achieve the aims and objectives of this project.

The objective of sensitivity analysis is to identify the most important factors in the model. It is then possible to fix the unimportant factors which reduces the number of uncertain inputs and would not lead to a significant reduction in output uncertainty. Thus, enabling the ability to simplify the model. Reducing the output uncertainty from its initial value to a lower pre-established threshold value as part of the assessment study gives the ability to narrow input uncertainties. Identifying the important inputs in a specific domain of output variables, for example, which combination of factors produce output values above or below given threshold. Sensitivity analysis has many methodologies, for instance, local sensitivity analysis, global sensitivity analysis and under global is variable sensitivity analysis, which will all be assessed as possible methods.

Underlying computational models may be too complex to evaluate, thus surrogate modelling is required to bypass this. The surrogate model is an approximation of the original computational model and once the surrogate model has been selected, parameter fitted based on info contained in the experimental design and associated runs of the original computational model, the accuracy of the surrogate model shall be estimated by some kind of validation technique.

2.2 Modelling

Zero-D models assume a uniform distribution of the fundamental parameters with any of the organs and vessels of the model at any instant in time, whilst dimensional models of higher order are able to consider the variation of these parameters within the body's organs and its vessels. They use a set of simultaneous ordinary differential equations (ODE's). They feature major components of the system and are good for examining the global distribution of pressure, flow and volume. When the dimensions of the model increase, they begin to use a series of partial differential equations. 1D models represent the wave transmission effects within the vessels, 2D models can represent the radial variation of velocity in an axisymmetric tube and 3D models compute complex flow patterns [6].

2.2.1 Zero-D Models

Zero-D models, lumped parameter models, are simplified mathematical representation of a real system that treat the system's components as discrete entities without spatial dimensions. In cardiovascular models, the blood flow in the circulatory system and electric conduction are similar, such that blood pressure and flow rate are represented by voltage and current, as Poiseuille's law is the equivalent to Kirchoffs current law and Navier-Stokes equation is similar to Ohm's Law. Zero-D models began with the Windkessel models (two-element, three-element) [6].

Windkessel Models

The two-element Windkessel model has resistance and compliance in it. Resistance is calculated as:

$$R = \frac{(P_{\text{ao,mean}} - P_{\text{ven,mean}})}{\text{CO}} \approx \frac{P_{\text{ao,mean}}}{\text{CO}} \tag{2.1}$$

where $P_{\text{ao,mean}}$ and $P_{\text{ven,mean}}$ is the mean aortic and venous pressure and CO is the cardiac output. Compliance is determined by the elasticity of the arteries. The value of total compliance is calculated as:

$$C = \frac{\Delta V}{\Delta P} \tag{2.2}$$

where it is a ratio of change in volume and change in pressure.

Two-element Windkessel pressure goes done exponentially with the time constant measuring how quickly the pressure goes down in the arteries during diastole [7].

$$\frac{dP}{dt} = \frac{1}{C}\left(Q_{\text{in}} - \frac{P}{R}\right) \tag{2.3}$$

This is how the ordinary differential equation looks like for the two-element Windkessel model.

The two-element Windkessel has only aortic pressure. Characteristic impedance was added to the two-element model which reflects the wave propagation properties of the arterial system, improving its accuracy at modelling high-frequency pressure and flow interactions.

This model adds characteristic impedance (Z):

$$\frac{dP}{dt} = \frac{1}{C}\left(Q_{\text{in}} - Q_{\text{out}}\right) \tag{2.4}$$

where:

$$Q_{\text{out}} = \frac{P}{R+Z} \tag{2.5}$$

Three-element to four-element was made to reduce the errors in low frequency range as the three-element model cannot compute the global inertial effects of the entire arterial system. So a fourth element was added, total arterial inertance, it accounts for mass and inertia of the blood moving through the entire arterial system [7].

This model includes inertance (L) to account for global arterial inertia:

$$L\frac{dQ_{\text{in}}}{dt} + Q_{\text{in}}R + \frac{P}{C} = 0 \tag{2.6}$$

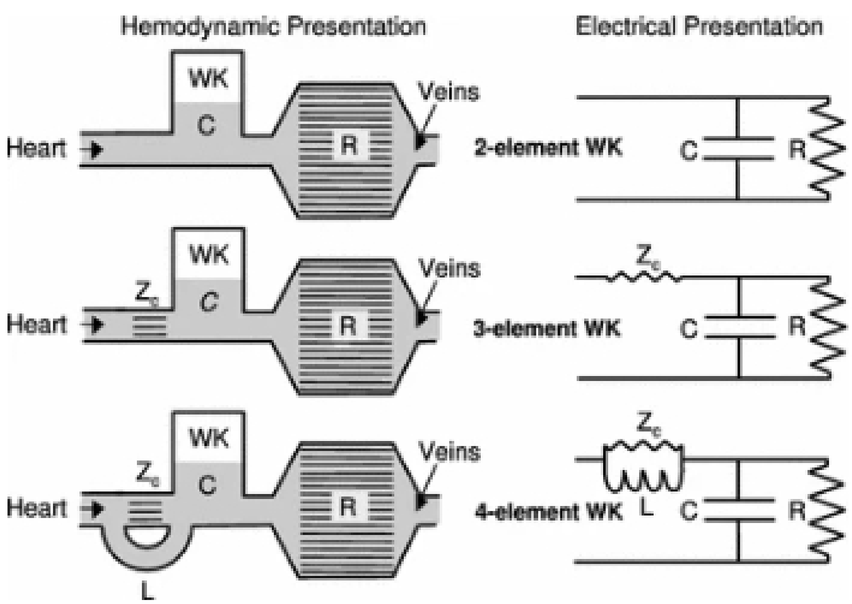

Figure 2.1: Figure 2.1 from Meyer et al. [1] illustrates... representation of the Windkessel models, showing how they increase in complexity by the addition of more elements.

Summary

The Windkessel models has clinical uses, such as prediction of cardiovascular risk as total arterial compliance plays a crucial role in finding pulse pressure and the model has parameters such as compliance, resistance and impedance to help in understanding some conditions such as hypertension. Simplicity and physiological understanding enables it to be used in clinical circumstances.

2.2.2 One-D Models

1-D models are based from the continuity and momentum equations, for conservation of mass and momentum in the arterial system, which factor for blood flow velocity, vessel cross sectional area and wall elasticity [8].

The models simulate the transmission and improved boundary conditions for 3D local models, capable of capturing systemic wave reflection effects [6]. The general assumption is that the pressure changes in the radial direction of the blood vessel dictates that the pressure is constant over any cross-section and that on integrating the conservation of momentum along the axis over the cross-sectional area of the vessel, the radial velocity terms are subsumed into an area term, so the flow velocity and pressure are treated as averaged quantities over the vessel's cross-section. The assumptions provide an accurate approximation of flow and pressure wave propagation in large arteries, where flow patterns are predominantly axial and radial effects are minor.

1-D models have mostly been applied to study the pulse wave transmission dynamics in arterial segments. Parker and Joans implementation of the 1-D model utilises the 1-D flow equations to describe the propagation of pressure and flow waves in arteries [9]. They assume uniform pressure, making them computationally efficient and well-suited for analysing wave intensity. Wave intensity analysis is used to quantify the energy and power carried by waves at some parts of the circulation, it is a 1-D analytical technique as it is based on the method of characteristics solutions of the 1-d equations derived from the conservation of mass and momentum in elastic vessels. Parker and Joans use this fact to analyse the waves in large system arteries and coronary arteries [10].

Summary

By assuming uniform pressure and treating flow velocity and pressure as averaged quantities, 1-D models has a balance between computational efficiency and physiological realism. The models are useful for studying wave dynamics due to their reliance on the conservation of mass and momentum equations, which create the interaction between arterial elasticity, flow and pressure.

2.2.3 Two-D and Three-D Models

2D and 3D models are mostly based on the Navier-Stokes equations. They are non-linear partial differential equations [6]. They can provide highly detailed and localised information, such as: pressure distributions within the vessel, which highlights areas of high stress, and velocity profiles of blood flow at any point in the vessel, which is important for understanding behaviour like turbulence.

However the computational demand is too high, they require solving complex equations, like Navier-Stokes equations, that describe fluid motion. To run the simulation will take significant time, hence why the scope of this project cannot consider Two-D and Three-D models as there is a need to find balance between efficiency and accuracy to be used in clinical settings.

2.2.4 Summary

Cardiovascular modelling has many variety and each come with their advantages and limitations. Zero-D models simplify the cardiovascular system into lumped parameters, allowing them to be ideal for understanding global phenomena. They are computationally efficient and used for assessing cardiovascular risks and conditions like hypertension. One-D models solve continuity and momentum equations which incorporate wave propagation dynamics. They create a balance in computational efficiency and physiological accuracy, making them useful for pulse wave transmission in large arteries. 2-D and 3-D models are much more localised as they are enable to view pressure distributions, velocity profiles and turbulence. However their computational cost limit their usage and thus cannot be considered when doing this project. The Zero-D model will be the main focus of this project as it is more useful in viewing the overall behaviour of the cardiovascular system and are computationally lighter than one-d models, as they only need to solve small number of ordinary differential equations. Although it is more inaccurate in comparison to One-d, they are more appropriate in regards to the goal of this projects, which concerns cardiovascular risks.

2.3 Sensitivity Analysis

The main purpose of sensitivity analysis is to view how uncertainty in the output of a model can be attributed to different sources of uncertainty in a model's input. Sensitivity analysis allows the study of which input variables contribute most to output behaviour, which input variable are non-influential and any interaction effects within the model. It can calculate multiple output values, where they can depend on large number of input parameters and some input variables maybe unknown, unspecified or defined with large imprecision range. There are two types of sensitivity analysis, local and global sensitivity analysis, and the latter has many variations to it, such as variational sensitivity analysis.

The objective of sensitivity analysis is factor prioritisation setting, factor fixing setting, variance cutting setting and factor mapping setting. The details are described in the "Handbook of Uncertainty Quantification (Ghanem et al., 2017, pp. 1106)" [11]. Factor prioritisation setting is the identification of the most important factors, which, if fixed, would lead too reduction in the uncertainty of the output. The factor fixing setting is the reduction of the number of uncertain input by fixing unimportant factors which, if fixed to any, would not lead to a significant reduction in output uncertainty. This can be the preliminary step before model input calibration. Variance cutting setting is the reduction in the output uncertainty from its initial value to a lower pre-established threshold value, as part of a risk assessment study. The factor mapping setting is the identification of important inputs in a specific domain of output values, for example, which combination of factors produces output values above or below the given threshold. In the scope of this project, sensitivity analysis is utilised to assess the contribution of the parameters that partake in forming the model of the heart.

The general model relationship for sensitivity analysis is given as follows [11].

$$Y = G(x) \tag{2.7}$$

where:

- Y: The model output.
- $G(x)$: The model function.
- $x = [x_1, x_2, \ldots, x_n]$: A vector of input parameters.

In the Windkessel model, the output (Y) can be represented as a function of the input parameters (x) through a model function ($G(x)$)[1]:

$$Y = G(x)$$

where:

- Y: Represents the output variable (e.g. blood pressure).
- x: Represents the input parameters (e.g. resistance R, compliance C, and cardiac output CO).
- $G(x)$: Describes the relationship between Y and x.

For resistance (R), the relationship is given as:

$$R = \frac{(P_{\text{ao,mean}} - P_{\text{ven,mean}})}{\text{CO}} \approx \frac{P_{\text{ao,mean}}}{\text{CO}}$$

where:

- $P_{\text{ao,mean}}$: Mean aortic pressure.
- $P_{\text{ven,mean}}$: Mean venous pressure.
- CO: Cardiac output.

For compliance (C), the relationship is defined as:

$$C = \frac{\Delta V}{\Delta P}$$

where:

- ΔV: Change in blood volume.
- ΔP: Change in pressure.

Applying the $Y = G(x)$ framework:

- The resistance formula shows how changes in $P_{\text{ao,mean}}$, $P_{\text{ven,mean}}$, and CO influence resistance (R).
- The compliance formula demonstrates the sensitivity of compliance (C) to changes in volume (ΔV) and pressure (ΔP).

2.3.1 Local Sensitivity

Local sensitivity analysis focuses on the study of the impact of small input variations around nominal values or the input of the model. Calculations or estimations of partial derivatives of the model at a specific point in the input variable space. Local methods do have limitations; however, they assume linearity and normality and they only consider local variations around a specific point rather than the entire domain of possible input parameter variations.

Previous studies of local sensitivity analysis for simplified models were useful for getting results with a fast run-time [12]. However, they failed to estimate global and multi-factorial sensitivity. For clinical measurements and reducing the parameters space, finding the key parameters is very important especially for the personalisation stage.

Mathematical Formulation

A system is described by a state vector $\mathbf{x}(t) \in \mathbb{R}^n$, governed by [11]:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \boldsymbol{\theta}, t)$$

where:

- $\mathbf{x}(t)$ are the state variables (e.g., x, y, z in the Lorenz system),
- $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_m]$ is the vector of model parameters,
- f is the vector field describing the system dynamics.

The local sensitivity of the i-th state variable with respect to parameter θ_j is defined as the partial derivative:

$$S_{ij}(t) = \frac{\partial x_i(t)}{\partial \theta_j}$$

This represents how sensitively $x_i(t)$ responds to small changes in θ_j at time t.

2.3.2 Global Sensitivity

Global sensitivity analysis is the statistical framework that overcomes the limitations of local methods. It differs from local sensitivity analysis, as it focuses on how small variations in an input around a given value change the output. Global sensitivity analysis considers numerical models in the entire domain of possible input parameter variations.

Global sensitivity is used to study mathematical models as a whole rather than just one of its solutions around a specific parameter value.

In previous study (by E. Karabelas et al.[13]) that used global sensitivity analysis to identify and quantify key parameters on the outputs of a four-chamber heart model. They used a surrogate model to approximate the computationally expensive. The global sensitivity analysis resulted in strong influence of left and right atrial pressure on model outputs. The benefit was that it was computationally efficient and it was able to be analysed holistically. However, there are limitations to this study, such as the surrogate model as some output features were excluded due to poor predictive performance and the surrogate still required around 7700 hours of high performance computing resources.

2.3.3 Variational Sensitivity Analysis

Define an output called the response function, the sensitivity to input variations around a nominal value can be studied using derivative information. The main issue of variational sensitivity analysis is to provide an efficient way to compute the gradient.

The aim of variational sensitivity analysis is to provide sensitivity information using derivatives, and the derivative of a function at a given point gives growth rate at that point. The larger the derivative, the more sensitive the parameter.

Variational sensitivity analysis is local when it focuses on input variations around a nominal set of input parameters. This approach assumes linearity in the relationship between inputs and outputs near the operating point.

Global variational sensitivity analysis uses the interactions and non-linearities in the parameter space, providing a more comprehensive understanding of parameter influence. It is closely related to the research domain called data assimilation. It consists of adjusting input parameters of model so that the system state fits given set of observations. Variational data assimilation translates this into optimal control problem whose aim is to minimise model observation misfit. Minimisation is performed using descent methods and gradient is computed using so called adjoint methods, which leverages the linearised version of the model equations, making it computationally efficient for high-dimensional systems. As stated in in the Handbook of Uncertainty Quantification (Ghanem et al., 2017, pp. 1124).[11]

Variational sensitivity analysis focuses on methods to evaluate sensitivities locally. The derivative that quantifies how small changes in input parameters affect the output is first defined, then adjoint method is shown to provide a powerful way to compute the derivatives. A brief overview about practical derivative computation is given and finally stability analysis, which examines how small perturbations to the system evolve over time, is mentioned, which is closely related to sensitivity analysis. s[14]

Morris' Method

This method is a global sensitivity analysis technique which is designed to identify important parameters, non-linear effects where parameter's influence changes depending on its value and interactions where a parameter's effect depends on other parameters.

Mathematically, for a model $Y = G(X)$ with input parameters $X = [x_1, x_2, \dots, x_k]$, the elementary effect (EE) for parameter x_i is calculated as [14]:

$$EE = \frac{f(\theta + \Delta) - f(\theta)}{\Delta}, \quad \Delta = \frac{l}{2(l-1)}, \tag{2.8}$$

where:

- Δ: A step size that determines how much x_i is perturbed.
- $f(\theta)$: The model output and θ is the input parameter.

Sobol Analysis

Sobol indices is used to show how much of the variability in model output is dependent upon each of the input parameters. They help to identify which inputs are most influential and whether interactions between inputs significantly affect the output. However, Sobol analysis is not intended to identify the cause of the input variability, it just indicates what impact and to what extent it will have on model output. Sobol does not assume between model input and output, it evaluates the full range of each input parameter variation between parameters, but it has high computation intensity being the main drawback. Therefore, the sample size needed for performing a reliable Sobol sensitivity analysis depends on complexity and number of parameters being evaluated. The most sensitive parameters should have narrow confidence intervals.

In a thesis by (R. Gul et al.[15]), the Sobol indices was applied to perform global sensitivity analysis, helping quantify variations in input parameters that would affect model output. In the model, Sobol indices identifies and ranks the most critical parameters contributing to the output uncertainty, it then decomposes the total variance of the output into contributions, next it pinpoints which parameters influences the outputs the most and the model help prioritise calibration using clinical measurements, then the Sobol analysis highlights how interactions between parameters affect outputs.

Issues arise when the method is constrained by computational demands, reliance on input distributions and challenges in high-dimensional spaces. It is considered the gold standard in terms of sensitivity analysis.

2.3.4 Summary

Sensitivity analysis has been applied to identify key parameters, improving computational efficiency and clinical relevance. There are challenges for computational demands, reliance on input distributions and handling high-dimensional parameters spaces. Local methods are computational efficient but limited, global methods provide greater insight but the computational cost is high. We go further with global sensitivity analysis, with Morris' method and Sobol analysis, but requires consideration of computational resources and parameter complexity.

2.4 Surrogate Modelling

When modelling, the model may be too computationally expensive to evaluate, the surrogate model was built to bypass this fact. The surrogate model is an approximation of original computational model:

$$x \in D_X \subset \mathbb{R}^d \mapsto y = G(x)$$

which is constructed based on a limited number of runs of the true model, the so-called experimental design:

$$\mathcal{X} = \{x^{(1)}, \ldots, x^{(n)}\}.$$

The parameters have to be fitted based on the information contained in the experimental design $\mathcal{X}$ and associated runs of the original computational model $\mathcal{Y} = \{y_i = G(x^{(i)}), i = 1, \ldots, n\}$. The accuracy of the surrogate would be estimated by some kind of validation technique. x is a vector of input variables which belongs to domain $\mathcal{X}$. The domain is a subset of $\mathbb{R}^d$ in the space of d-dimensional real numbers. $y = G(x)$ represents the original complex computational model. As stated in the Handbook of Uncertainty Quantification (Ghanem et al., 2017, pp. 1290-1) [11].

Surrogate modelling enables efficient sensitivity analysis by approximating the outputs of interest with far fewer computations and they provide efficient means to integrate uncertainties and quantify output variability. In this circumstance, the cardiovascular model is being reduced computationally by forming a surrogate as described in Chapter 1. Methods to surrogate modelling involve polynomial chaos expansion and Kriging.

2.4.1 Polynomial Chaos Expansion (PCE)

The main idea of PCE is to express the model output as a spectral expansion of orthogonal polynomials with the polynomials chosen to match the probability distributions of the input parameters. It is used in uncertainty quantification for a model and sensitivity analysis where the PCE coefficients provide variance-based sensitivity measures, such as Sobol indices. Once the surrogate is built, the evaluation run-time is very fast.

However, the number of terms in the expansion grows rapidly with the number of inputs variables, making it challenging for high-d problems.

In a study conducted by Huberts et al.[16], they use PCE to enhance the personalisation of a cardiovascular pulse wave propagation model. It was used to identify which parameters impacts the output variance the most and determining which parameters has minimal impact on the output variance.

They constructed a surrogate model of the pulse wave propagation model, approximating the behaviour and using polynomial basis functions dependent on stochastic input parameters. The new model allows a reduction in the need for direct evaluations of the full computational model which is computationally expensive. There are 73 stochastic input parameters, representing uncertainties in patient-specific and physiological conditions and for this case, PCE was used to study the mean brachial flow and systolic pressure. The study found that PCE reduces computational cost significantly, where they compare it to Satellite's Monte Carlo method and the PCE provided accurate estimates for sensitivity indices and parameter rankings for parameter prioritization and fixing.

However, it is too computationally expensive during high-order expansions, when the polynomial basis expands rapidly. The PCE assumes independent input parameters and uniformly distributed uncertainties, which may not always align with real-world clinical data as with correlated inputs, the correlations must be known, whereas these correlations are not known in a clinical setting.

2.4.2 Kriging

Kriging provides a predictive model for a system based on a set of observed data points and is an unbiased estimation method, which can not only predict the response values of unknown samples but also predict their uncertainty [17]. Kriging also provides an estimate of the uncertainty in these predictions. Kriging predicts the observed outputs at the training points and then provides a confidence interval for its predictions, which is useful for understanding the reliability of the surrogate model. It provides precise predictions, especially for smooth functions. An advantage of the Kriging model is its ability to reduce the number of needed parameters if the database is small. However, as Kriging involves inverting the covariance matrix, which has a cubic computational complexity, thus making it computationally expensive for large datasets.

In a study by J.Chen et al.[18], Kriging was used as a statistical method for modelling and optimising mechanical properties of 3-D printed metamaterials designed to mimic the aortic tissue. Kriging provided a prediction for stress-strain curves for untested metamaterial designs based on learned input-output relationships and estimates of uncertainty.

However the limitations of this method is that the Kriging model assumes that the covariance between output levels can be separated from the covariance between input functions. This simplifies computation, but does not fully capture complex dependencies between parts of the stress-strain curve.

2.4.3 Summary

When sensitivity analysis is performed, the computational expense increases, especially with the usage of global methods, such as Sobol and Morris' method. Surrogate modelling enables efficiency with exploring input-output relationships and uncertainty quantification whilst reducing the computational cost and maintaining reasonable accuracy. Both surrogate modelling methods are very effective, however during the literature survey, Kriging tends to not be used, whereas PCE was used the large majority of the time for the cardiovascular system as it provides explicit, analytical expressions for outputs as functions of inputs and Kriging requires additional steps like variance decomposition, making it less direct and interpretable than PCE for this purpose, where PCE provides Sobol indices and global sensitivity metrics.

Chapter 3

Approach

3.1 Aims

As detailed in Chapter 1, this project aims to recreate the heart using a Zero-D model and to perform sensitivity analysis on the model and then to create a surrogate model to improve the computational efficiency of the model. The end product aims to be computationally fast enough to be used for clinical purposes.

3.2 Objectives

Table 3.1: Project Requirements Overview

Component	**Description**
Heart Model	Develop a computational model of the heart using a Zero-D model. Implement both single- and two-chamber models (LV and LA), dictated by differential-algebraic equations (DAEs).
Sensitivity Analysis (Model)	Evaluate parameter influence using global sensitivity techniques. Sobol sensitivity analysis is selected for its accuracy and compatibility with surrogate modelling.
Surrogate Model	Construct a surrogate to reduce computational cost. Polynomial Chaos Expansion (PCE) is used to approximate model outputs, leveraging OpenTURNS for automatic basis construction and normalisation.
Sensitivity Analysis (Surrogate)	Re-perform Sobol sensitivity analysis using the surrogate model to compare performance, stability, and accuracy against the full model.

3.2.1 Modelling

In the current scope of the research of this area, researchers use mechanistic models for the model of the heart, choosing from Zero-D to 3-D model. Zero-D model will be the model of choice for this project. With all things considered regarding computational cost and accuracy, the 2-D and 3-D models cannot be chose due to their computational cost, which would not be usable in clinical settings, where results are required within a limited time frame. The Zero-D model is great at generalising the cardiovascular system and is computationally extremely fast and is accurate enough to provide sufficient results, hence why it is chosen to be used over One-D models. This project aims to maximise the efficiency of the model so it can be used quickly, especially for emergency settings. The model is defined as a differential algebraic equation (DAE), Windkessel model, with 9 parameters dictating the elastance timing, valve resistances, compliances, and pressure-volume relationships and for the two chamber model, 22 parameters were used for the left atrium and the left ventricle, which includes the same from the left ventricle only model but has the parameters for the left atrium and arterial system. For the model to be formed, the libraries used are the Assimulo's IDA solver (SUNDIALS) to simulate the stiff DAE system. The heartbeats are irregular, mimicking real physiological variability via heart rate variability (HRV).

3.2.2 Sensitivity Analysis

This project will aim to maximise the accuracy available from getting good results from sensitivity analysis. At the moment, the best method is Sobol sensitivity analysis in comparison to morris' method and local sensitivity due to its accuracy and efficiency, which also works well with surrogate modelling, thus implementing it . The library to be used is SALib, which is used for the original model and to the surrogate model, the results produced are the first order Sobol indices and the total effect Sobol indices, which are visualised using heat maps. The sampling method for the sensitivity analysis used is Saltelli through SALib.

3.2.3 Surrogate Model

Surrogate modelling aims to reduce computational cost of the already designed model. There are two surrogate model techniques used in the review for the literature survey, polynomial chaos expansion (PCE) and Kriging. This project will be using the former, as the latter will be too computational expensive and the aims of this project is to be computationally effective to be used in clinical settings. The PCE is utilised via a library called OpenTURNS as it works well with the sensitivity analysis as it is possible to automatically normalise the polynomial expansion for Sobol decomposition. The alternative option was Chaospy which also does PCE, however in order to do sensitivity analysis, you need to manually build the orthogonal polynomials and need to normalise the inputs. This automatic process allows OpenTURNS to be implemented more easily with the model's need for sensitivity analysis. During the literature review, surrogate models have not been implemented on Zero-D models.

3.2.4 Conclusion

There is an attempt to maximise the balance between computational efficiency and accuracy by the choices of the method. Once the models have been built and the surrogate model has been made, sensitivity analysis can be applied to both and comparisons can be made in terms of the difference in accuracy of results and run time. The results will be compared to current implementations of the same methods to the methods implemented in this report.

Chapter 4

Experiments

4.1 The Model Implementation

To simulate cardiovascular dynamics and analyse the effects of parameter uncertainty, this project implements two lumped-parameter models of the heart: a single-chamber model (left ventricle only) and the two-chamber model (left ventricle + left atrium). These models capture the essential behaviour using pressure, flow, and volume variables, along with time-varying elastance to represent contraction and relaxation. Dictated by differential-algebraic equations (DAEs), both models incorporate physiologically relevant features such as heart rate variability (HRV), valve dynamics, and systemic circulation. The elastance approach defines stiffness as a function of time within each heartbeat, enabling realistic simulation of cardiac cycles and allows for sensitivity and surrogate analysis.

4.1.1 One Chamber Implementation:

The single-chamber model focuses solely on the left ventricle, capturing the core dynamics of pressure and volume during a cardiac cycle. It uses a time-varying elastance function to describe systolic and diastolic phases. The model is dictated by a set of DAEs that simulate the interaction between the ventricle, systemic circulation, and valves. The realistic physiological nature is maintained by using parameters such as ventricular elastance, arterial compliance, and valve resistances. The model includes an event-handling mechanism and stochastic heartbeat intervals via HRV, allowing for the simulation of beat-to-beat variation seen in real physiology.

State Variables

The model state variables are:

- P_{LV}: Left ventricular pressure
- P_{sa}: Systemic arterial pressure

- P_{sv}: Systemic venous pressure
- V_{LV}: Left ventricular volume
- Q_{av}: Aortic valve flow
- Q_{mv}: Mitral valve flow
- Q_s: Systemic flow

Model Parameters

Table 4.1: Cardiovascular model parameters for one chamber

Parameter	**Description**	**Value**
τ_{es} (s)	Elastance end-systole timing	0.30
τ_{ep} (s)	Elastance end-phase timing	0.45
R_{mv} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Mitral valve resistance	0.06
Z_{ao} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Aortic impedance	0.033
R_s $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Systemic resistance	1.11
C_{sa} $(\frac{\text{ml}}{\text{mmHg}})$	Systemic arterial compliance	1.13
C_{sv} $(\frac{\text{ml}}{\text{mmHg}})$	Systemic venous compliance	11.0
$E_{\max}$ $(\frac{\text{mmHg}}{\text{ml}})$	Maximum elastance	1.50
$E_{\min}$ $(\frac{\text{mmHg}}{\text{ml}})$	Minimum elastance	0.03

The parameters used for the project is the same as the data used in "Personalised parameter estimation of the cardiovascular system: Leveraging data assimilation and sensitivity analysis" by Harry Saxton [19] as it is the standardised parameters used for the development of the cardiovascular model.

Time-Varying Elastance

Define cycle time within beat: $t_i = (t - t_r) \mod \tau$ which represent the normalised time within the current cardiac cycle.

Definition of $t_i = (t - t_r) \mod \tau$

This formula defines the time within the current cardiac cycle:

- t – Current simulation time.
- t_r – Reference time of the most recent cycle.
- τ – Duration of the current cycle.

- t_i – Elapsed time within the current cycle, ranging from 0 to τ. Resets every beat to ensure elastance dynamics follow the correct phase of contraction and relaxation.

Elastance Function

This elastance is the time-varying stiffness of the left ventricle. Its a function of time that shows how the heart contracts and relaxes the during each cardiac cycle.

$E(t) = E_{\min} + (E_{\max} - E_{\min})E_p(t)$

Where:

$$E_p(t) = \begin{cases} \frac{1}{2}\left(1 - \cos\left(\pi \frac{t_i}{\tau_{es}}\right)\right), & t_i \le \tau_{es} \\ \frac{1}{2}\left(1 + \cos\left(\pi \frac{t_i - \tau_{es}}{\tau_{ep} - \tau_{es}}\right)\right), & \tau_{es} < t_i \le \tau_{ep} \\ 0, & t_i > \tau_{ep} \end{cases}$$

The $t_i \le \tau_{es}$, $\tau_{es} < t_i \le \tau_{ep}$ and $t_i > \tau_{ep}$ describe the different phases of the heart. The first phase is of the heart contracting (systole), the second condition is the relaxing phase after contraction (diastole) and the last condition is the resting phase where the heart is completely relaxed and the blood is filling the ventricle. The valve acts as a regulator for pressure within the left ventricle, where once a certain pressure is reached, the valve opens to allow the contraction to occur.

Elastance Derivative

The time derivative of $E(t)$ is a DAE that needs to be solved. It is needed as $E(t)$ is not constant, which varies within each heartbeat. When modelling the dynamics of P_{LV}, the effect of $\frac{dE_p(t)}{dt}$ must be included to involve the rate of change of pressure.

$$\frac{dE(t)}{dt} = (E_{\max} - E_{\min}) \cdot \frac{dE_p(t)}{dt}$$

Where:

$$\frac{dE_p(t)}{dt} = \begin{cases} \frac{\pi}{2\tau_{es}} \sin\left(\pi \frac{t_i}{\tau_{es}}\right), & t_i \le \tau_{es} \\ -\frac{\pi}{2(\tau_{ep} - \tau_{es})} \sin\left(\pi \frac{t_i - \tau_{es}}{\tau_{ep} - \tau_{es}}\right), & \tau_{es} < t_i \le \tau_{ep} \\ 0, & t_i > \tau_{ep} \end{cases}$$

Valve Flow Equations

$$Q_{av} = \begin{cases} \frac{P_{LV} - P_{sa}}{Z_{ao}}, & \text{if } P_{LV} > P_{sa} \\ 0, & \text{otherwise} \end{cases}$$

When the ventricular pressure is greater than the arterial pressure, the blood flows out the heart through the aorta.

$$Q_{mv} = \begin{cases} \frac{P_{sv} - P_{LV}}{R_{mv}}, & \text{if } P_{sv} > P_{LV} \\ 0, & \text{otherwise} \end{cases}$$

When the blood flows into the left ventricle through the mitral valve only if venous pressure is greater than ventricular pressure. Otherwise, the blood does not flow in.

Differential-Algebraic Equations (DAEs)

- $P_{LV} = (Qmv - Q_{av})E(t) + \frac{p_{LV}}{E(t)}\frac{dE(t)}{dt}$
- $P_{sa} = \frac{Qav - Q_s}{C_{sa}}$
- $P_{sv} = \frac{Q_s - Qmv}{C_{sv}}$
- $V_{LV} = Qmv - Q_{av}$
- $Q_{av} = \frac{P_{LV} - P_{sa}}{Z_{ao}}$ if $P_{LV} > P_{sa}$, else 0
- $Q_{mv} = \frac{P_{sv} - P_{LV}}{R_{mv}}$ if $P_{sv} > P_{LV}$, else 0
- $Q_s = \frac{P_{sa} - P_{sv}}{R_s}$

Event Trigger

An event is triggered when: $(t - t_r) \mod \tau = \tau_{es}$

Heart Rate Variability (HRV)

The HRV function stochastically generates cardiac cycle times $\tau_i \in [0.4, 1.1]$ seconds:

$$\tau_i \sim \mathcal{U}(0.4, 1.1)$$

$$t_{\tau_{i+1}} = t_{\tau_i} + \tau_i$$

This defines the duration of each heartbeat, allowing the model to simulate heart rate variability.

Implementation of Cardiovascular Simulation

The model is executed using the `Assimulo` package with the IDA solver from the SUNDIALS suite, which is well-suited for stiff systems and DAE problems.

The class `CardiovascularModel` implements the system's dynamics via a residual function `res()`, defining the DAE system. These are implemented in `ShiElastance()` and `DShiElastance()` respectively.

The model tracks cardiac events using `root()`, which runs when the current time in cycle $t_i = (t - t_r) \mod \tau$ reaches τ_{es} (end-systolic time). Upon an event, `handle_event()` updates internal cycle counters and switches the phase label between systole and diastole. It also updates the current beat duration τ based on the heart rate variability sequence.

Initial conditions for pressures, flow, and volume are assigned, and the DAE is solved using `IDA`. The solver is configured with tight tolerances (10^{-6}), a maximum step size of 0.05 s, and is run over the full cardiac sequence generated by `HRV()`. The model is initialised with default parameters values. The left ventricular pressure (p_{LV}), systemic arterial pressure (p_{sa}), and systemic venous pressure (p_{sv}) are plotted alongside left ventricular volume (V_{LV}) over the 33–35 s window.

Finally, the function returns the input features X_{train} (parameter matrix), output trajectories Y_{train} (selected outputs such as P_{LV}, P_{sa}, V_{LV}), and time stamps for downstream use, including surrogate model training and sensitivity analysis.

4.1.2 Two Chamber Implementation (Left Ventricle and Left Atrium):

The two-chamber model is introduced that includes both the left ventricle and the left atrium. This model captures the dynamics between atrial and ventricular chambers and their interactions with the systemic circulation. Each chamber is modelled using a time-varying elastance function with distinct systolic and diastolic timings. State variables and parameters are introduced to represent volumes and pressures in both chambers, proximal and distal systemic arteries, veins, and valve flow dynamics. This allows for more detailed analysis of cardiac function and provides a robust framework for sensitivity and surrogate modelling.

State Variables

The model state variables are:

- P_{LV}: Left ventricular pressure
- P_{LA}: Left atrial pressure
- V_{LV}: Left ventricular volume
- V_{LA}: Left atrial volume
- P_{sas}: Systemic arterial pressure
- Q_{sas}: Flow in systemic arterial
- P_{sat}: Systemic arterial pressure
- Q_{sat}: Flow in systemic arterial
- P_{svn}: Systemic venous pressure
- Q_{svn}: Flow from systemic veins to atrium

- Q_{av}: Aortic valve flow
- Q_{mv}: Mitral valve flow

Model Parameters

Table 4.2: Cardiovascular model parameters (from code implementation)

Parameter	Description	Value
$v_{0,lv}$(ml)	Unstressed LV volume	10.00
$E_{\min,lv}$ $(\frac{\text{mmHg}}{\text{ml}})$	LV minimum elastance	0.10
$E_{\max,lv}$ $(\frac{\text{mmHg}}{\text{ml}})$	LV maximum elastance	2.50
$\tau_{es,lv}$(s)	LV systolic duration	0.30
$\tau_{ed,lv}$(s)	LV relaxation duration	0.45
$v_{0,la}$ (ml)	Unstressed LA volume	10.00
$E_{\min,la}$ $(\frac{\text{mmHg}}{\text{ml}})$	LA minimum elastance	0.15
$E_{\max,la}$ $(\frac{\text{mmHg}}{\text{ml}})$	LA maximum elastance	0.25
$\tau_{es,la}$(s)	LA systolic duration	0.045
$\tau_{ed,la}$(s)	LA relaxation duration	0.09
Z_{ao} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Aortic valve impedance	0.033
R_{mv} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Mitral valve resistance	0.060
C_{sas} $(\frac{\text{ml}}{\text{mmHg}})$	Aortic root compliance	0.08
R_{sas} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Aortic root resistance	0.003
L_{sas} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Aortic root inertance	6.2×10^{-5}
C_{sat} $(\frac{\text{ml}}{\text{mmHg}})$	Systemic arterial compliance	1.60
R_{sat} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Systemic arterial resistance	0.05
L_{sat} $(\frac{\text{mmHg}\cdot\text{s}^2}{\text{ml}})$	Systemic arterial inertance	0.0017
R_{sar} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Arteriole resistance	0.50
R_{scp} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Capillary resistance	0.52
C_{svn} $(\frac{\text{ml}}{\text{mmHg}})$	Systemic venous compliance	20.5
R_{svn} $(\frac{\text{mmHg}\cdot\text{s}}{\text{ml}})$	Systemic venous resistance	0.075

The parameters in the two chamber model is derived from “Convergence, sampling and total order estimator effects on parameter orthogonality in global sensitivity analysis” by Harry Saxton et al.[3] but were changed to optimise the shape and represent the two chamber model.

Elastance Function - Left Ventricle

$$E_{LV}(t) = E_{\min,lv} + (E_{\max,lv} - E_{\min,lv}) \cdot E_{p,lv}(t)$$

$$E_{P_{LV}}(t) = \begin{cases} \frac{1}{2}\left(1 - \cos\left(\pi \frac{t_i}{\tau_{es,lv}}\right)\right), & t_i \leq \tau_{es,lv} \\ \frac{1}{2}\left(1 + \cos\left(\pi \frac{t_i - \tau_{es,lv}}{\tau_{ep,lv} - \tau_{es,lv}}\right)\right), & \tau_{es,lv} < t_i \leq \tau_{ep,lv} \\ 0, & t_i > \tau_{ep,lv} \end{cases}$$

Elastance Derivative - Left Ventricle

$$\frac{dE_{LV}(t)}{dt} = (E_{\max,lv} - E_{\min,lv}) \cdot \frac{dE_{p,lv}(t)}{dt}$$

$$\frac{dE_{p,lv}(t)}{dt} = \begin{cases} \frac{\pi}{2\tau_{es,lv}} \sin\left(\pi \frac{t_i}{\tau_{es,lv}}\right), & t_i \leq \tau_{es,lv} \\ -\frac{\pi}{2(\tau_{ep,lv} - \tau_{es,lv})} \sin\left(\pi \frac{t_i - \tau_{es,lv}}{\tau_{ep,lv} - \tau_{es,lv}}\right), & \tau_{es,lv} < t_i \leq \tau_{ep,lv} \\ 0, & t_i > \tau_{ep,lv} \end{cases}$$

The formulas for the elastance function of the left ventricle and its derivative is the same as before, there is no change to the formulas of the left ventricle.

Elastance Function — Left Atrium

$$E_{LA}(t) = E_{\min,la} + (E_{\max,la} - E_{\min,la}) \cdot E_{p,la}(t)$$

$$E_{p,la}(t) = \begin{cases} \frac{1}{2}\left(1 - \cos\left(\pi \frac{t_i}{\tau_{es,la}}\right)\right), & t_i \leq \tau_{es,la} \\ \frac{1}{2}\left(1 + \cos\left(\pi \frac{t_i - \tau_{es,la}}{\tau_{ep,la} - \tau_{es,la}}\right)\right), & \tau_{es,la} < t_i \leq \tau_{ep,la} \\ 0, & t_i > \tau_{ep,la} \end{cases}$$

Elastance Derivative — Left Atrium

$$\frac{dE_{LA}(t)}{dt} = (E_{\max,la} - E_{\min,la}) \cdot \frac{dE_{p,la}(t)}{dt}$$

$$\frac{dE_{p,la}(t)}{dt} = \begin{cases} \frac{\pi}{2\tau_{es,la}} \sin\left(\pi \frac{t_i}{\tau_{es,la}}\right), & t_i \leq \tau_{es,la} \\ -\frac{\pi}{2(\tau_{ep,la} - \tau_{es,la})} \sin\left(\pi \frac{t_i - \tau_{es,la}}{\tau_{ep,la} - \tau_{es,la}}\right), & \tau_{es,la} < t_i \leq \tau_{ep,la} \\ 0, & t_i > \tau_{ep,la} \end{cases}$$

The concept of the conditions is the same as the left ventricle but the phases occur at different times, hence why they are defined differently. The left atrium receives oxygenated blood from the lungs and pumps the blood only to the left ventricle, after a certain pressure level is reached, hence why the valve opens.

The two chamber model also implements the idea of the Heart Rate Variability (HRV) and then event trigger mechanism.

Two-Chamber Model Implementation

The methodology of implementing the two chamber model is very similar in comparison to the one chamber model where it is different by the number of parameters used. There are 22 parameters as opposed to 9 as it involves the left atrium as well as the left ventricle.

The simulation is launched in the `run_minimal()` routine, where initial conditions and physiologically-informed parameter values are set for elastances, compliances, resistances, and inertances of the circulatory segments. State variables include pressures (P_{LV}, P_{LA}, P_{sas}, P_{sat}, P_{svn}), volumes (V_{LV}, V_{LA}), and flows across systemic and valvular components. The algebraic variables Q_{av} and Q_{mv} are explicitly marked in `algvar` to ensure proper DAE indexing.

The simulation is configured for a 35-second interval, with results saved at 5000 evenly spaced time points. Outputs are extracted and visualised as pressure, volume, and flow curves over a narrowed 33–35 s window. These plots allow assessment of the beat-to-beat interactions between atrial and ventricular chambers, systemic perfusion, and valve responses. The model is deterministic with a 1-second cardiac cycle, suitable for sensitivity analysis or surrogate modelling.

4.1.3 Sensitivity Analysis

Two types of Sobol indices are computed: first-order and total-effect.

First-order Sobol Index

$$S_i = \frac{\mathrm{Cov}\left[f(\mathbf{A}), f\left(\mathbf{A}_B^{(i)}\right)\right]}{\mathrm{Var}[f(\mathbf{X})]} \tag{4.1}$$

Where:

- $\mathbf{A}$ and $\mathbf{B}$ are two independent $N \times d$ input sample matrices.
- $\mathbf{A}_B^{(i)}$ is the matrix $\mathbf{A}$ with the i-th column replaced by the i-th column of $\mathbf{B}$.
- $f(\cdot)$ is the scalar-valued model output.

Total-effect Sobol Index

$$S_{T_i} = \frac{\mathbb{E}\left[f(\mathbf{A})^2 - f(\mathbf{A})f\left(\mathbf{A}_B^{(i)}\right)\right]}{\mathrm{Var}[f(\mathbf{X})]} \tag{4.2}$$

These indices quantify, respectively:

- S_i: the main effect of input X_i on the output variance.
- S_{T_i}: the total contribution of X_i to the variance, including interactions.

Saltelli Sampling

Saltelli sampling is a variance-based global sensitivity analysis technique. Initially to sample using Saltelli's method, generate two independent matrices of input samples.

$$\mathbf{A} \in \mathbb{R}^{N\times d}, \quad \mathbf{B} \in \mathbb{R}^{N\times d} \tag{4.3}$$

For each variable $i \in \{1, \dots, d\}$, a hybrid matrix is made by replacing the i_{th} column of $\mathbf{A}$ with the same column from matrix $\mathbf{B}$. It allows for efficient computation for first order and total order Sobol indices, computing $N \cdot (d+2)$ model evaluations [20]. This is described in the formula below:

$$\left[\mathbf{A}_B^{(i)}\right]_{:,j} = \begin{cases} \mathbf{B}_{:,j}, & \text{if } j = i \\ \mathbf{A}_{:,j}, & \text{otherwise} \end{cases}$$

Saltelli's sampling method can be used to compute sensitivity indices in order to compare against the PC surrogate-based approaches.

$$\hat{S}_i = \frac{1}{\widehat{V} f(\lambda)} \left[\frac{1}{M} \sum_{m=1}^{M} f\big(\lambda(\boldsymbol{\xi}^{(m)})\big) \Big(f\big(\lambda(\boldsymbol{\xi}_{-i}^{(m)})\big) - f\big(\lambda(\boldsymbol{\xi}^{(m)})\big) \Big) \right] \tag{4.4}$$

Where:

- $\hat{S}_i$: Estimated first-order Sobol' index for input X_i
- $\widehat{V} f(\lambda)$: Estimated variance of the model output
- $f(\lambda(\boldsymbol{\xi}^{(m)}))$: Model evaluated at the m-th sample from base matrix $\mathbf{A}$
- $\boldsymbol{\xi}_{-i}^{(m)}$: Hybrid sample where only the i-th component is taken from matrix $\mathbf{B}$ (i.e., $\mathbf{A}_B^{(i)}$)
- M: Number of Monte Carlo samples

Table 4.3: Correspondence between Sobol mathematics and code implementation

Concept	**Mathematical Expression**	**Code Equivalent**
First-order Sobol index	$S_i = \frac{\mathrm{Cov}[f(\mathbf{A}), f(\mathbf{A}_B^{(i)})]}{\mathrm{Var}[f(\mathbf{X})]}$	`sobol.analyze(..., calc_second_order=False)['S1']`
Total-effect Sobol index	$S_{T_i} = \frac{\mathbb{E}[f(\mathbf{A})^2 - f(\mathbf{A}) f(\mathbf{A}_B^{(i)})]}{\mathrm{Var}[f(\mathbf{X})]}$	`sobol.analyze(..., calc_second_order=False)['ST']`
Hybrid matrix construction	$\mathbf{A}_B^{(i)}$	`saltelli.sample(...)` auto-generates these
Variance denominator	$\mathrm{Var}[f(\mathbf{X})]$	Implicit in SALib via sample matrices

4.1.4 Surrogate Modelling

Polynomial Chaos Expansion (PCE)

The concept of PCE is that the model output $Y = G(X)$ is expressed as a series, where X are the random variables in the system. The aim is to expand Y in a basis of orthogonal polynomials that match the distribution of the input variables.

Model as a series

The model output $Y = G(X)$ is described as an infinite series:

$$Y = \sum_{j=0}^{\infty} y_j Z_j \tag{4.5}$$

Where:

- Y: the output quantity (random variable)
- y_i: deterministic coefficients
- Z_j: orthogonal polynomial basis functions

Polynomial Chaos Basis

$\psi_j(X)$, polynomial chaos basis functions, is formed from products of univariate orthogonal polynomials. The basis $\psi_j(X)$ is formed as such:

$$\psi_j(X) = \prod_{k=1}^{d} P_{\alpha_k}^{(k)}(X_k)$$

where:

$P_{\alpha_k}^{(k)}(X_k)$ is a univariate orthogonal polynomial of degree α_k, tailored to the distribution of X_k.

$$\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$$

is the multi-index associated with basis function ψ_j.

The basis is chosen such that:

$$\langle \psi_i, \psi_j \rangle = \delta_{ij} \tag{4.6}$$

In this implementation, Legendre polynomials are used, which are orthogonal over uniform distributions. It is consistent with the uniform parameter bounds, which enables for optimal basis function behaviour. OpenTURNS automatically constructs this orthonormal basis and handles multi-dimensional indexing internally.

Truncation

The series is truncated as it is infinite, doing total-degree truncation up to order p.

$$|\alpha| = \sum_{i=1}^{d} \alpha_i \leq p$$

This leads to a total of:

$$\mathrm{card}(A_{d,p}) = \binom{d+p}{p}$$

The multivariate polynomial basis terms that exist up to total degree p in d variables are counted. p dictates the accuracy and complexity of the surrogate. Higher values increase representational power but also require exponentially more samples. When $d = 9$ parameters and $p = 3$, the basis size is:

$$\mathrm{card}(A_{9,3}) = \binom{9+3}{3} = 220$$

Computation of Coefficients

The coefficients are computed, y_j. The goal is to find all the constants of y_j to build an accurate surrogate model. The coefficients y_j are fitted using regression-based approaches (via OpenTURNS' FunctionalChaosAlgorithm).

Implementation of Polynomial Chaos Expansion (PCE)

The PCE expresses the model as:

$$f(x) \approx \sum_{j=0}^{P-1} y_j \psi_j(x)$$

This representation is represented in code via: `meta(sc.transform(np.atleast_2d(x)))[0,0]`, where `sc` is a `StandardScaler` that standardizes the input vector x, and `meta` is the trained surrogate metamodel obtained from `FunctionalChaosAlgorithm`.

To construct the polynomial basis functions ψ_j, Legendre polynomials are used. They are orthogonal with respect to the uniform distribution of the input parameters. These are generated in code using: `ot.LegendreFactory()` from `OpenTURNS`, using this condition:

$$\langle \psi_i, \psi_j \rangle = \delta_{ij}$$

The total number of basis terms depends on the input dimension d and the total polynomial degree p, and is given by the combinatorial formula:

$$\mathrm{card}(A_{d,p}) = \binom{d+p}{p}$$

This is computed in code using `comb(d + p, p)`.

To truncate the infinite series, a total-degree condition is applied to the multi-index $\alpha = (\alpha_1, \ldots, \alpha_d)$ such that:

$$|\alpha| = \sum_{i=1}^{d} \alpha_i \leq p$$

The truncation is implemented using: `getBasisSizeFromTotalDegree(p)`.

Finally, the coefficients y_j in the expansion are fitted via regression using OpenTURNS' `FunctionalChaosAlgorithm(...).run()`, which performs a least-squares fit of the model response to the polynomial basis functions using the training data.

Basis Size Example

In the two-chamber cardiovascular model, the input dimension is $d = 22$. The number of PCE basis terms increases rapidly with the polynomial degree p:

Table 4.4: Number of basis terms for different polynomial degrees ($d = 22$)

Degree p	**Number of Basis Terms $\binom{d+p}{p}$**
2	276
3	2,300
4	14,950
5	80,730

The number of required training samples (N_{train}) should be at least 2–3 times the number of basis terms to ensure well-conditioned regression. This becomes computationally expensive for high-degree expansions.

Chapter 5

Results and Discussion

5.1 Evaluation Method

The different steps will be evaluated differently for this project. The modelling aspect will be evaluated on the shape they present and how it looks in comparison to the shapes provided on online models. The sensitivity analysis will be evaluated on the results they provide and the difference in the values presented by online implementations. The surrogate modelling cannot be evaluated against implementations made by others as they could not be found, therefore they will be evaluated on how they compare to the shape of the graphs of the original model.

5.2 Modelling

The one chamber and two chamber model is outputted showing the pressure and volume curves, showing the shape that describes it.

5.2.1 The One Chamber Model

The one chamber model is outputted and compared to the model from "New perspectives on sensitivity and identifiability analysis using the unscented kalman filter" by Harry Saxton et al. [2]. The evaluation of the model is done by simply viewing the shapes of the graph and if the values that the graphs peak at are physiologically sensible.

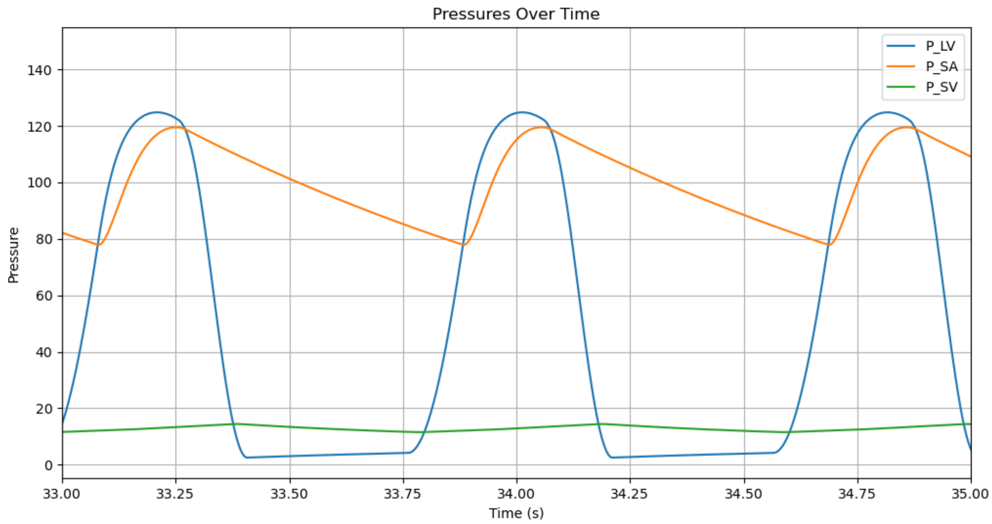

Figure 5.1: Pressure Dynamics Over Time for One Chamber

In Figure 5.1, the graph describes the pressures over time in a cardiovascular system, from a time frame of 33 to 35 seconds.

The blue line describes the left ventricular pressure, where it peaks at around 120mmHg. This peak corresponds to the contraction phase. The drop represents diastole, showing it relaxes and fills.

The orange line describes the systemic arterial pressure where it lags behind the left ventricular pressure curve. It peaks shortly after the left ventricle pressure, indicating that the blood is sent through to the aorta, however it decreases gradually.

The green line represents the systemic venous pressure where it is a very low curve with a range of 10 to 20mmHg. This shows the stability of the venous pressure, which is typical with a system of less pulsatile flow.

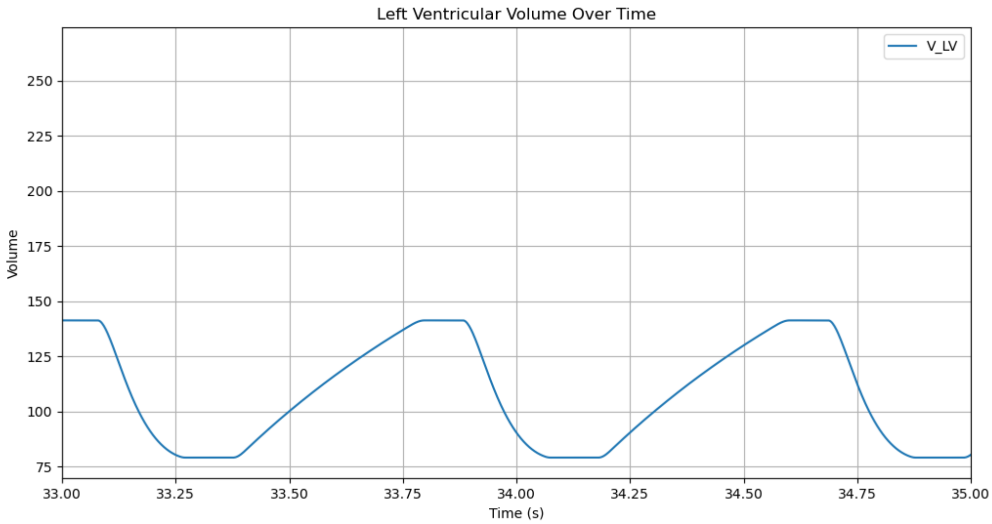

Figure 5.2: Left Ventricular Volume Dynamics Over Time

The Figure 5.2 provides the volume dynamics within the left ventricle. The waveform is periodic and it matches the cardiac cycle from Figure 5.1 as it drops sharply at the start, when pressure is high and increases as pressure is low.

The results provided shows the physiological relationships of pressure and volume over a 2 second window to show how the heart beat presents these dynamics within that time frame. In the study done by Harry Saxton et al.[2], the graphs show a similar shape but within in a different time frame.

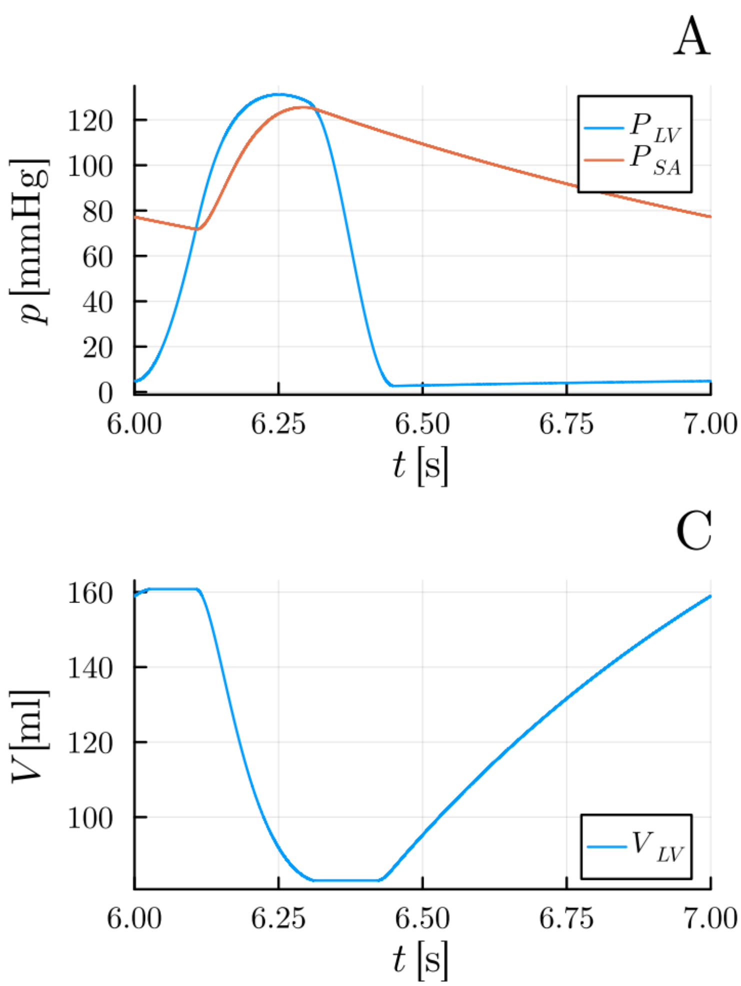

Figure 5.3: Left ventricular pressure and volume dynamics over time by Harry Saxton et al.[2]

The graphs in Figure 5.3 shapes are similar to the Figures in 5.1 and 5.2 even though they maximum and minimum of the peaks of both graphs are slightly different, they provide results that are physiologically accurate to a real cardiovascular system. The graphs show the sharp increase in gradient that relates to the systolic nature of the heart, contracting to send blood around the body, therefore pressure increases and volume decreases, overcoming aortic resistance. The diastolic nature of the heart is also shown, where the decrease in pressure and increase in volume is shown as the chamber relaxes.

5.2.2 The Two Chamber Model

The same methodology of evaluation done to the one chamber model will be applied to the two chamber model, whereby the shape of the graphs and the bounds of the maximum and minimum values peaked from the graphs will be viewed if they align with other implementations and if they make sense in a physiological sense.

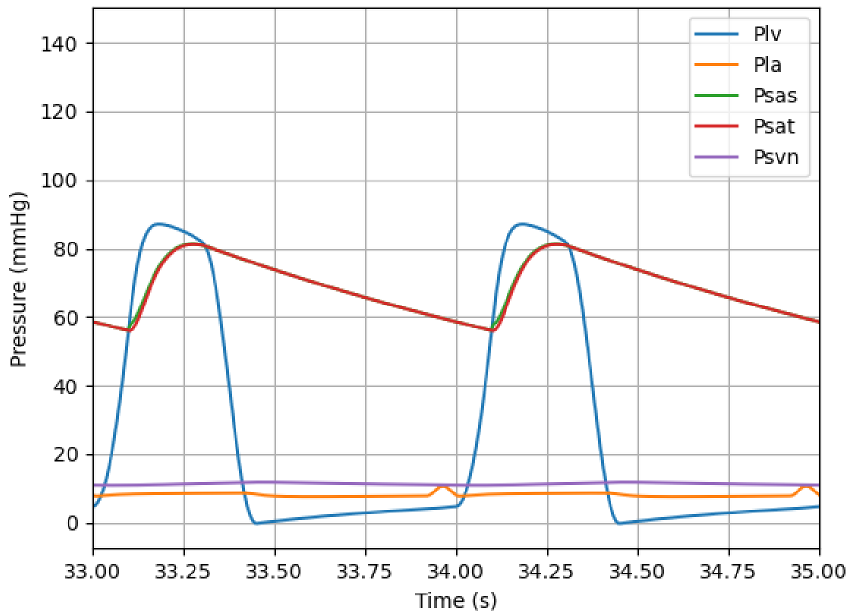

Figure 5.4: Pressure Dynamics Over Time for Two Chamber Model

In the Figure 5.4 the red line and green line represents the proximal systemic arterial pressure (P_{SAT}), systemic arterial trunk pressure (P_{SAS}). Both pressures between the systolic peaks show the elastic recoil of the arteries. The blue line is the left ventricle pressure, the orange line is the left atria pressure and the purple line is the venous pressure.

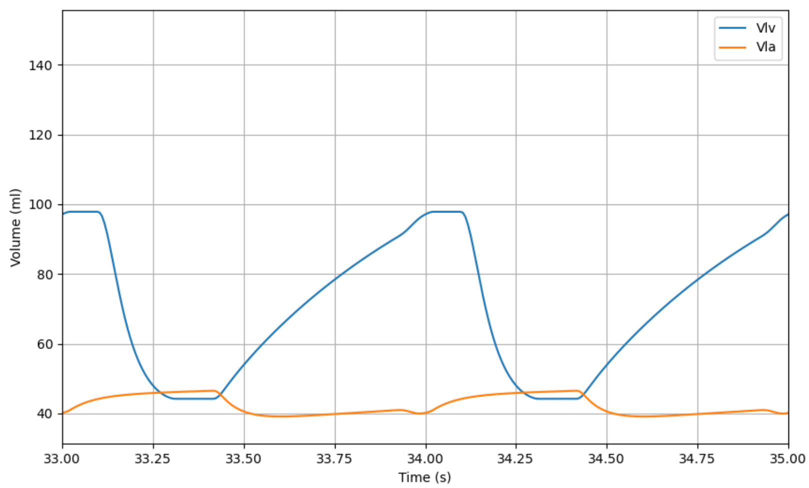

Figure 5.5: Volume Dynamics Over Time

In Figure 5.5, the blue line is the left ventricule volume and the orange line is the left atrium volume.

The Figures 5.4 and 5.5 give rise to similar shapes that are detailed in the one chamber model but at the peak of the shapes, the shapes vary slightly differently, showing that the two chamber model gives more detailed insight into the cardiovascular system. The two chamber model give a more realistic outcome in terms of its physiological nature due to the inclusion of the left atrium. The pressure curve has a slight bump to show what happens to the pressure just before systole occurs. The atrial pressure rises prior to ventricular systole, reflecting the kick that contributes to late diastolic filling. The volume waveform confirms that atrial influence has a tangible effect on ventricular preload and stroke volume as the atrial volume complements the atrial pressure and the ventricular volume complements the ventricular pressure.

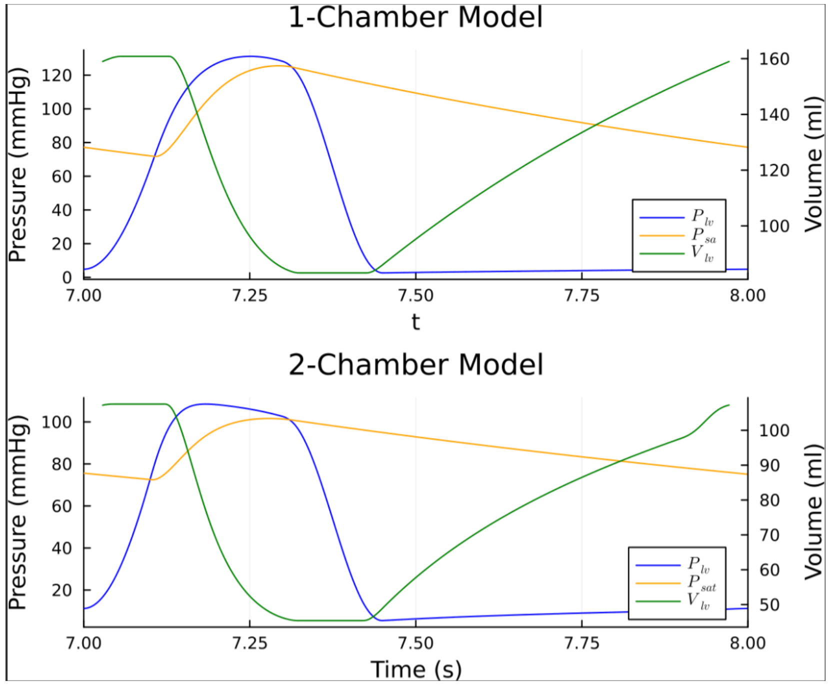

Figure 5.6: Left ventricular and Atrial pressure and volume dynamics over time by Harry Saxton et al.[3]

In the Figure 5.6 the shapes provided are similar to the graphs of 5.4 and 5.5. However, the peaks are at different values but the graphs provide results that makes sense physiologically due to the shapes they provide. They both capture the passive and active filling contributions with good accuracy.

5.3 Sensitivity Analysis

First and total order Sobol indices was implemented to the one and two chamber model, with low sample size and larger sample size to show the computational time and the accuracy of the results.

5.3.1 One Chamber Model Sensitivity Analysis

To understand the variations in input parameters that influence the behaviour of the one chamber cardiovascular model, a sensitivity analysis was conducted. This approach helps identify which parameters have the most significant effect on key outputs such as ventricular pressure and volume. The analysis provides insight into model robustness and highlights areas where precise parameter estimation is most critical. Multiple analysis was done at different sample points to establish the converging nature of the analysis and thus showing accuracy.

Table 5.1: Parameter bounds used in the simulation

Parameter	Lower Bound	Upper Bound
τ_{es} (s)	0.21	0.34
τ_{ep} (s)	0.36	0.50
R_{mv} ($\frac{\text{mmHg}\cdot\text{s}}{\text{ml}}$)	0.042	0.078
Z_{ao} ($\frac{\text{mmHg}\cdot\text{s}}{\text{ml}}$)	0.0231	0.0429
R_s ($\frac{\text{mmHg}\cdot\text{s}}{\text{ml}}$)	0.777	1.443
C_{sa} ($\frac{\text{ml}}{\text{mmHg}}$)	0.791	1.469
C_{sv} ($\frac{\text{ml}}{\text{mmHg}}$)	7.7	14.3
E_{max} ($\frac{\text{mmHg}}{\text{ml}}$)	1.05	1.95
E_{min} ($\frac{\text{mmHg}}{\text{ml}}$)	0.025	0.039

Low Sample Size

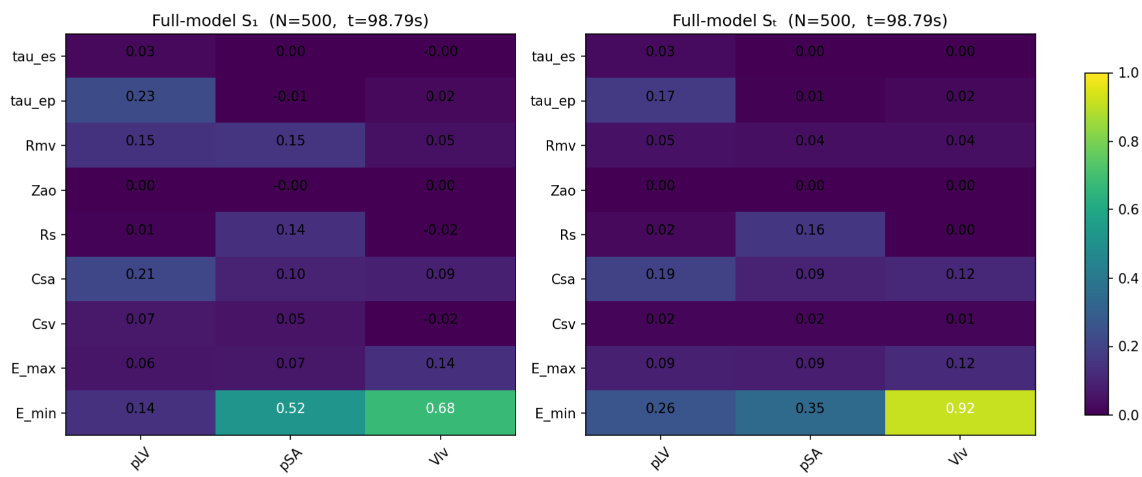

Figure 5.7: First Sobol Indices and Total Sobol Indices for 500 Samples

The heatmap in Figure 5.7 show the sensitivities of each parameter in the system, that would affect the P_{LV}, P_{SA}, V_{Lv}. The lighter colours symbolise the high effect on the system, and dark is for vice versa. This concept of the heatmaps will be applied across the other implementations. The graph titles show the run time of the models and the number of samples used.

The sensitivity analysis in Figure 5.7 shows for the outcomes; P_{LV}, P_{SA}, V_{Lv} for 9 parameters their contribution for each outcome. The analysis indicates E_{min} is the major contributing factor for both first and total Sobol indices, where total indices evaluates it at a greater effect. The evaluation is different but the general trend is the same as it sets the baseline stiffness of the heart chamber during diastole. C_{sa}, τ_{ep}, R_{mv} demonstrate moderate first-order effects, particularly for P_{LV} and P_{SA}. They relate to the damping and timing of pressure changes within the circulatory loop.

Sobol indices use variance and covariance for calculations of model outputs, thus requiring accurate estimation of conditional estimations. The estimators can have high variance, which would lead to noisy or unstable sensitivity analysis.

It is possible to increase accuracy and the best possible way to do so is by increasing the number of samples.

Greater Sample Size

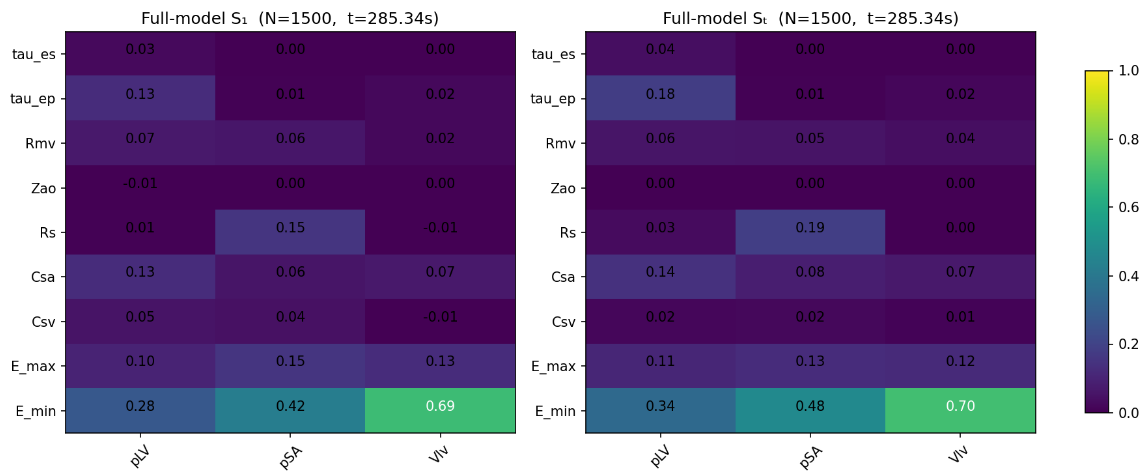

Figure 5.8: First Sobol Indices and Total Sobol Indices for 1500 Samples

In the sensitivity analysis with higher samples, Figure 5.8, the E_{min} stays as the dominant factor, thus showing that it is the most key driving factor amongst the other parameters. The magnitude decreased for the total Sobol indices evaluation from the 500 sample to the 1500 sample which is a good sign of variance redistribution becoming more accurate.

In comparison to the Figure 5.7, the other parameters have become more stable and showing more effect on the system. In Figure 5.7, some parameters had negative or near-zero S_1 values (e.g., $Z_{ao} = -0.01$). However, in Figure 5.8 the values are consistent and physically plausible. The computational time has increased by almost 200 seconds, and the problem will have to be addressed to resolve this.

5.3.2 Two Chamber Model Sensitivity Analysis

The technique to analyse the two chamber model is the same in regards to the one chamber model. There are 22 parameters to consider, therefore the computational time would be different and the effects on the model would be different. Table 5.2 shows the parameter bounds used for the two chamber sensitivity analysis for each parameter.

Table 5.2: Parameter Uncertainty Bounds for Two Chamber Model

Parameter	Lower Bound	Upper Bound
$v_{0,lv}$ (ml)	10.000	10.001
$E_{\min,lv}$ $\left(\frac{\text{mmHg}}{\text{ml}}\right)$	0.070	0.130
$E_{\max,lv}$ $\left(\frac{\text{mmHg}}{\text{ml}}\right)$	1.750	3.250
$\tau_{es,lv}$ (s)	0.210	0.340
$\tau_{ed,lv}$ (s)	0.360	0.585
$v_{0,la}$ (ml)	10.000	10.001
$E_{\min,la}$ $\left(\frac{\text{mmHg}}{\text{ml}}\right)$	0.105	0.175
$E_{\max,la}$ $\left(\frac{\text{mmHg}}{\text{ml}}\right)$	0.175	0.325
$\tau_{es,la}$ (s)	0.0315	0.0585
$\tau_{ed,la}$ (s)	0.0630	0.1170
Z_{ao} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.0231	0.0420
R_{mv} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.0420	0.0780
C_{sas} $\left(\frac{\text{ml}}{\text{mmHg}}\right)$	0.0560	0.1040
R_{sas} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.0021	0.0039
L_{sas} $\left(\frac{\text{mmHg·s}^2}{\text{ml}}\right)$	4.34×10^{-5}	8.06×10^{-5}
C_{sat} $\left(\frac{\text{ml}}{\text{mmHg}}\right)$	1.120	2.080
R_{sat} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.035	0.065
L_{sat} $\left(\frac{\text{mmHg·s}^2}{\text{ml}}\right)$	0.0019	0.00221
R_{sar} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.350	0.650
R_{scp} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.364	0.676
C_{svn} $\left(\frac{\text{ml}}{\text{mmHg}}\right)$	14.30	26.65
R_{svn} $\left(\frac{\text{mmHg·s}}{\text{ml}}\right)$	0.0525	0.0975

Low Sample Size

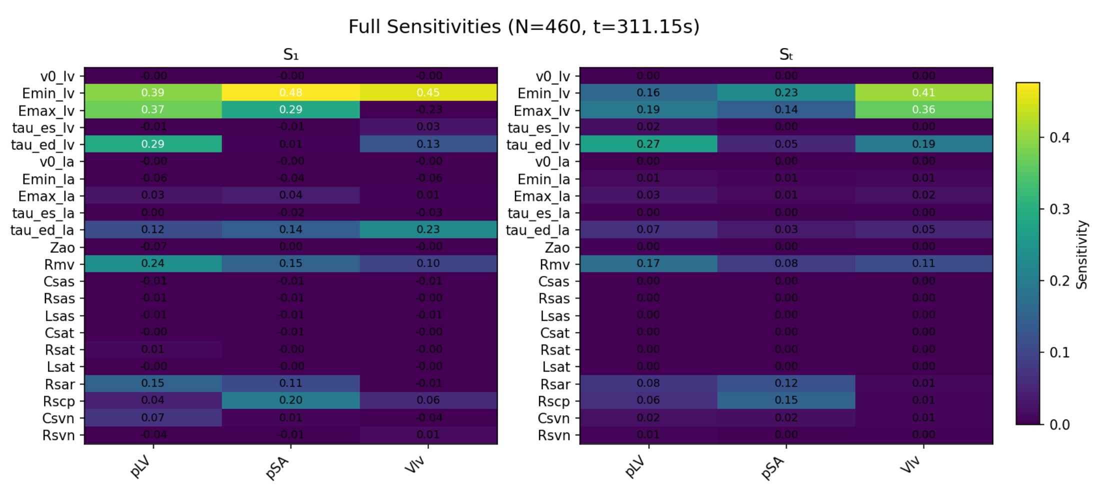

Figure 5.9: First Sobol Indices and Total Sobol Indices for 460 samples

In accordance to the first order Sobol indices sensitivity analysis in Figure 5.9, the $E_{min,lv}$ and $E_{max,lv}$ are both prominent features of the model. $\tau_{\text{ep,lv}}$ also contributes to V_{lv} and P_{LV}. Other parameters such as R_{mv}, R_{sar} and R_{scp} have some level of influence on the model as shown by the analysis. The trends tend to be the same for the total Sobol indices analysis but they are not as expressive as the first order analysis.

There are some negative values indicating noise within data which occurs due to the low sample size as some values may be unstable or unreliable.

Greater Sample Size

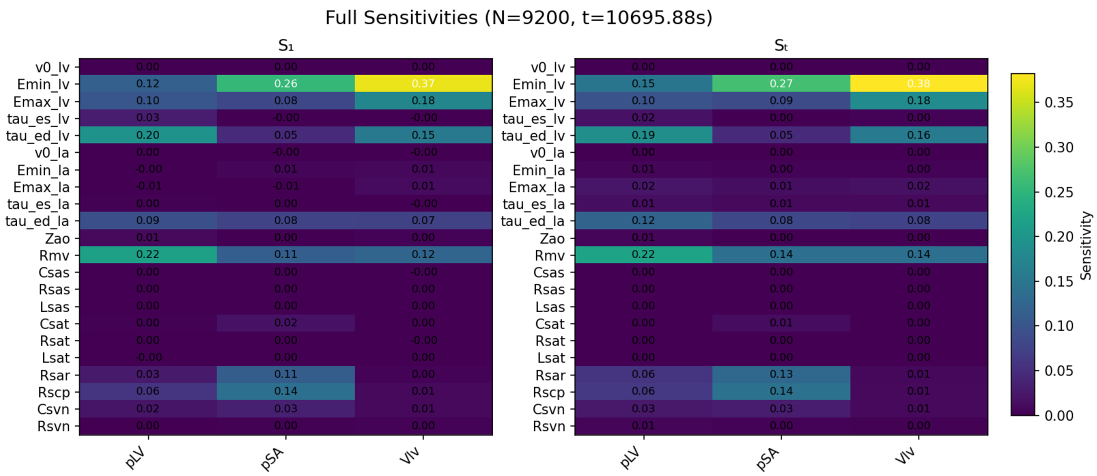

Figure 5.10: First Sobol Indices and Total Sobol Indices for 9200 samples

The high sampling size would show low variance in first order and total order Sobol indices. The differences between parameters are much more significant and the negative values and noise are more suppressed.

The same parameters show their significance to the model when observing Figure 5.9 but are expressed differently in Figure 5.10 especially in $E_{min,lv}$, where the influence in P_{LV} and P_{SA} is downplayed. A lot of the parameters are near zero influence, showing they are non-influential. The total order analysis remains very similar to the first order analysis indicating weak interaction effects.

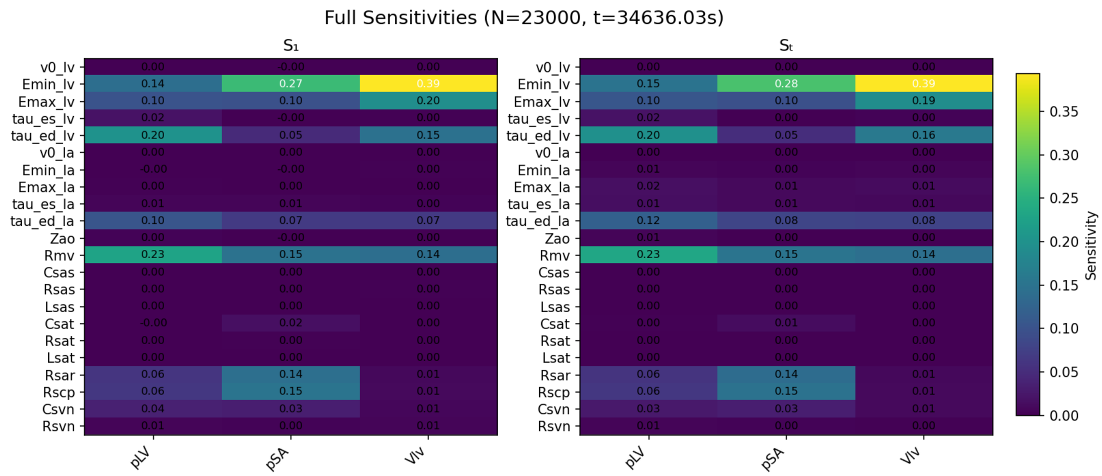

Figure 5.11: First Sobol Indices and Total Sobol Indices for 23000 samples

Another sample is ran but with much greater sampling size to decipher whether the effects of the previous results were accurate and the trends and results are almost identical. Thus, indicating the stability in the results when utilising greater sample size. The greater sample size is used to show the convergence in the results.

5.4 Surrogate Modelling

To accelerate sensitivity analysis on the time-dependent outputs of the cardiovascular model, a surrogate model was constructed using Polynomial Chaos Expansion (PCE). The surrogate approximates the model response over a fixed time window $[33, 35]$ seconds by learning a mapping from input parameters to compressed waveform representations.

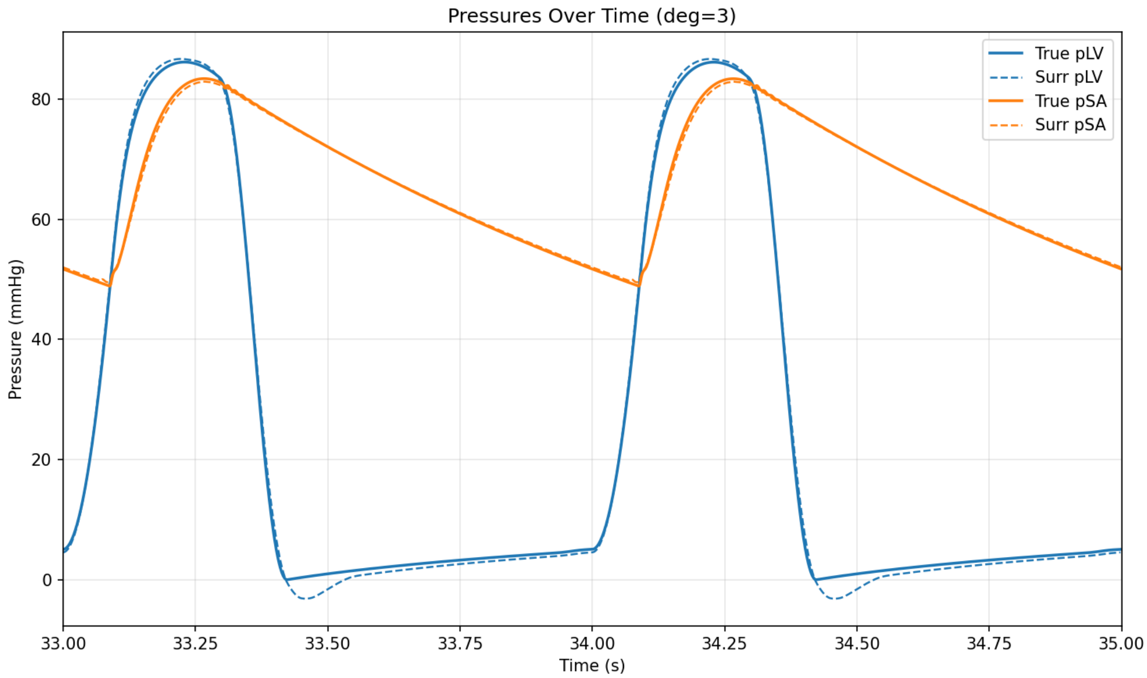

Figure 5.12: Surrogate model of one chamber

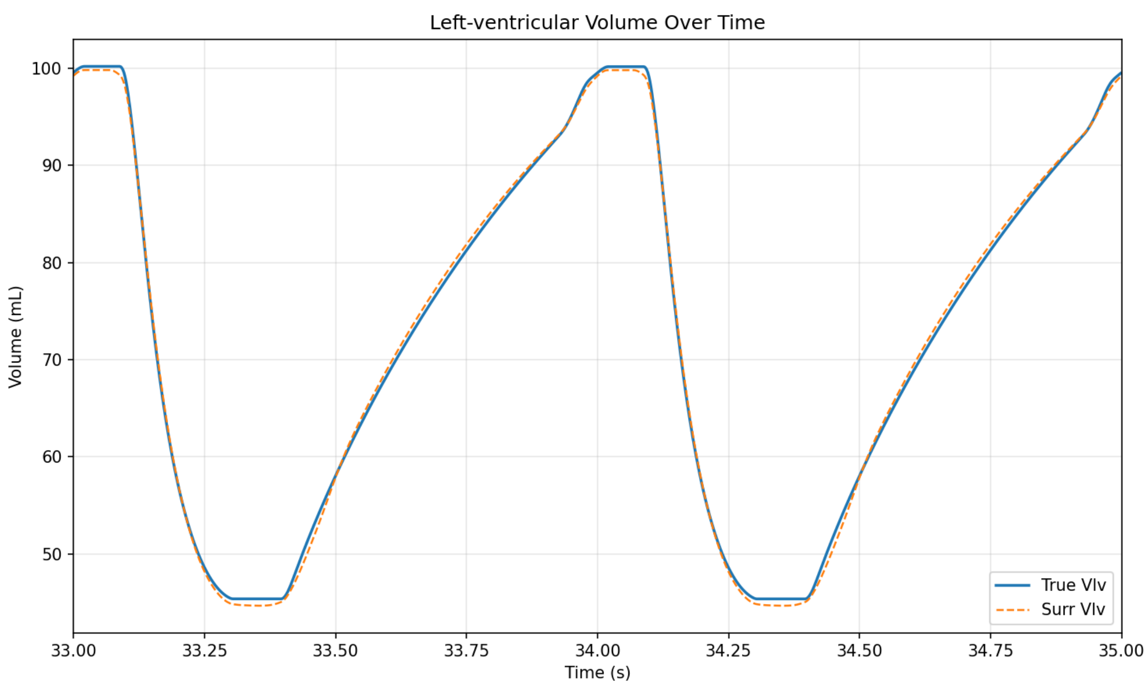

Figure 5.13: Surrogate model of one chamber

The surrogate model fits almost perfectly alongside the original model curves indicating the accuracy, as shown in Figures 5.12 and 5.13, even when a polynomial degree of 3 is used.

5.4.1 One Chamber Model Sensitivity Analysis

The one chamber model computationally reduced via polynomial chaos expansion and the aim is to maintain is to maintain the accuracy whilst reducing run time, for the purpose that it can be used in clinical settings.

Low Sample Size

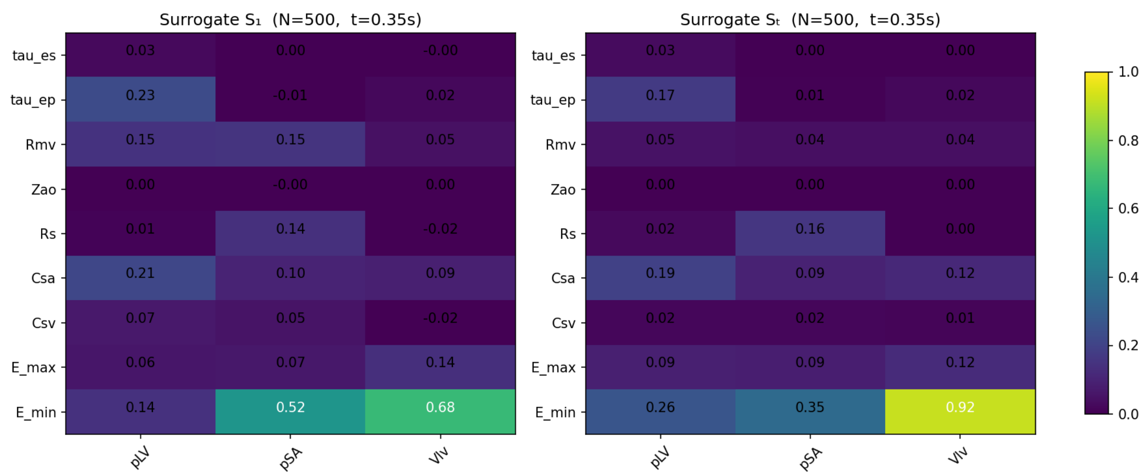

Figure 5.14: First Sobol Indices and Total Sobol Indices for 500 Samples for Surrogate Model

In Figure 5.14, the computational time has decreased from 98 seconds, as shown in Figure 5.7, to 0.35 seconds, as shown in Figure 5.14. The surrogate model almost maps to the original model exactly, where the individual values are almost the same. The polynomial order of the surrogate model is 5, which enables accurate calculations for samples of this size.

Greater Sample Size

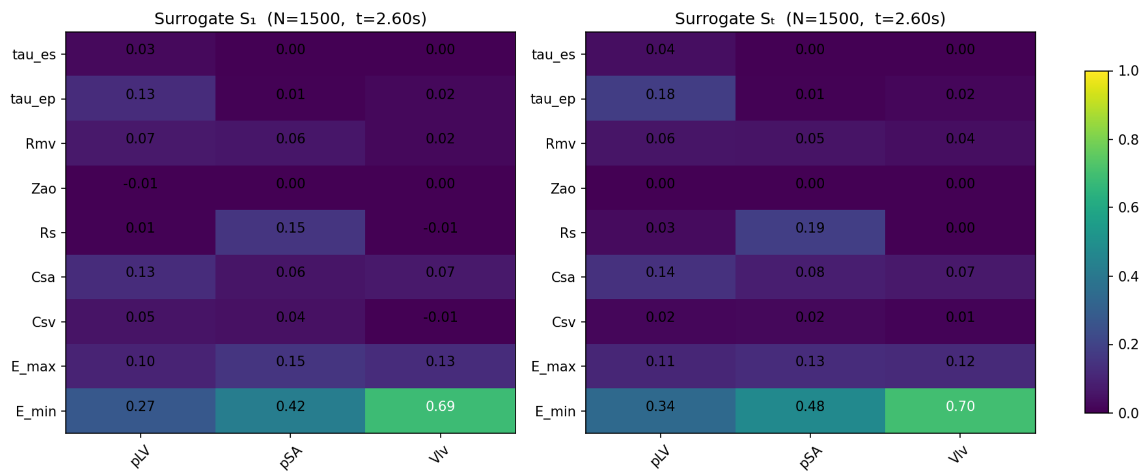

Figure 5.15: First Sobol Indices and Total Sobol Indices for 1500 Samples

In Figure 5.15, the surrogate Sobol analysis is near identical to the full model as shown in 5.8. $E_{\min}$ still dominates (e.g., $S_1 = 0.69$, $S_T = 0.70$ for Vlv), moderate influences from $E_{\max}$, R_s, and C_{sa} are preserved, weak or near-zero effects (e.g. Z_{ao} and C_{sv}) are similarly replicated. The surrogate captures interactive contributions of parameters.

The number of samples being 1500 means that the estimator variance drops, and for the surrogate model to match this means that the implementation must be robust and the functional forms used in the surrogate successfully approximates the mapping. This indicates the surrogate models ability to replicate the results from the original model. When comparing Figure 5.14, the computational time difference is 2.25 seconds but the accuracy is a lot higher when comparing it.

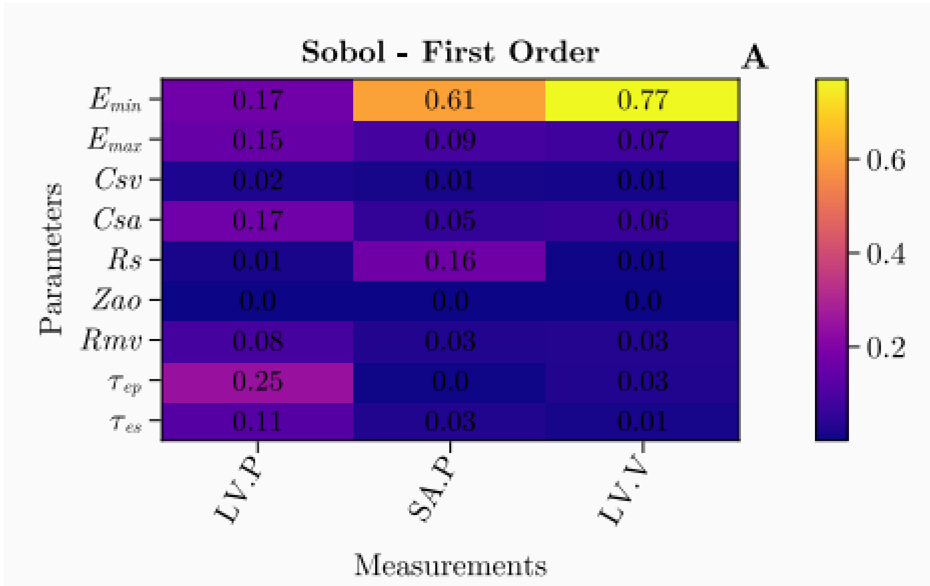

Figure 5.16: First Sobol Indices by Harry Saxton et al. [2]

When comparing the sensitivity analysis in Figure 5.16, the results are similar and the trends are the same, thus showing the physiological importance of the parameters shown above and the patterns they form regardless of the implementation method due to the realism of the physiological implementation.

5.4.2 Two Chamber Model Sensitivity Analysis

The procedure to implement sensitivity analysis to the two chamber model is similar to the methodology in the one chamber model. To simply apply the polynomial chaos expansion and to apply sensitivity analysis to it.

Low Sample Size

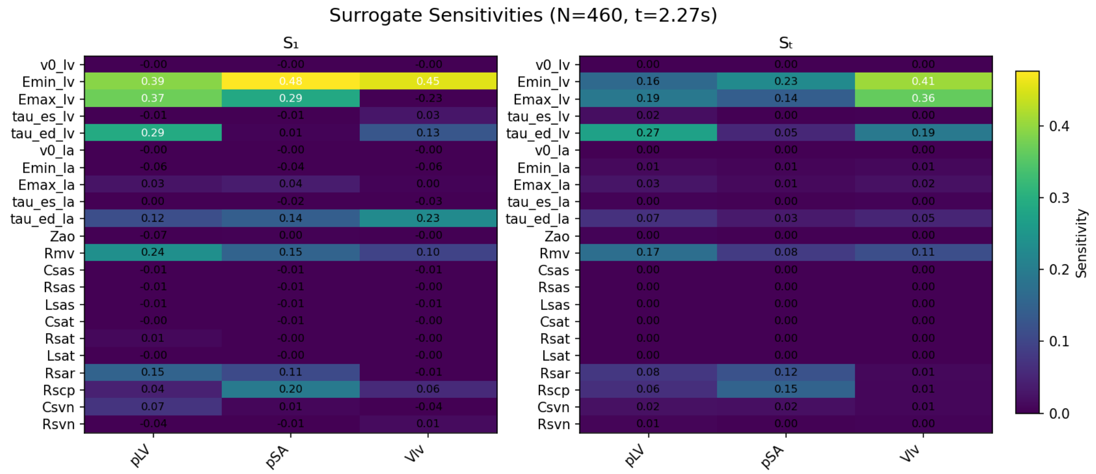

Figure 5.17: First and Total Sobol Indices for a surrogate model with Sample Size of 460

The computational time has decreased significantly, whereby in Figure 5.9, the time is 311.15 seconds but in this model it ran in 2.27 seconds. Whilst this drop in timing had occurred, the accuracy is maintained, indicating the surrogate's functionality being correct and accurate to the original model.

Greater Sample Size

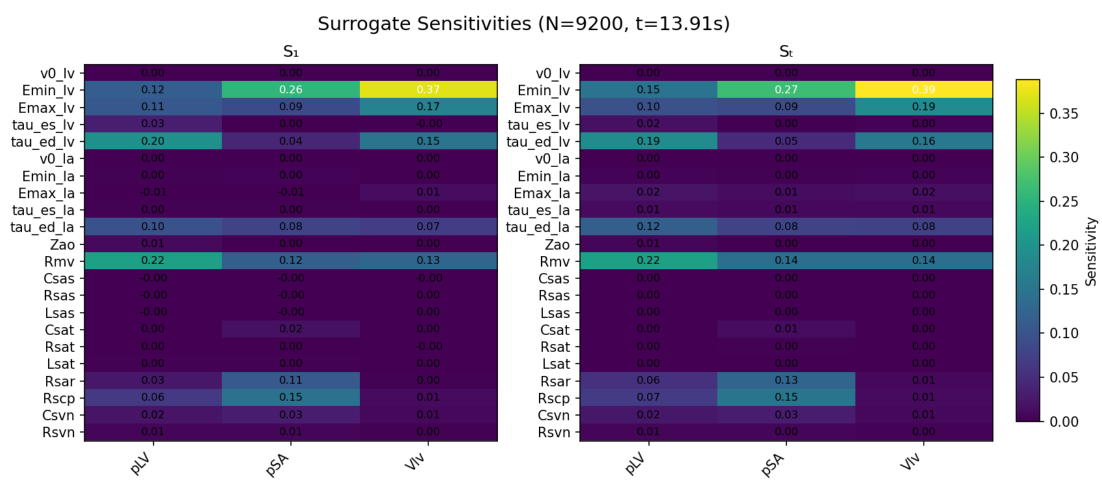

Figure 5.18: First and Total Sobol Indices for a surrogate model with Sample Size of 9200

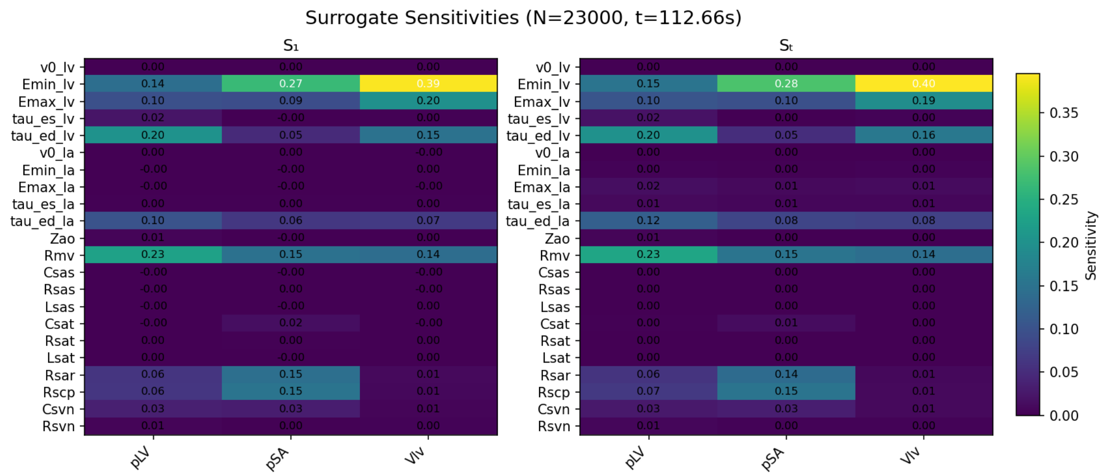

Figure 5.19: First and Total Sobol Indices for a surrogate model with Sample Size of 23000

For both Figures 5.18 and 5.19, the computational time has decreased significantly. In Figure 5.10, the computational time recorded is 10695.88 seconds but the recorded time for the surrogate model with the same sample size is 13.91 seconds and for Figure 5.11 the run time is 34636.03 seconds, approximately 9.6 hours, but the surrogate model run time for the same sample size is 112.66 seconds. This improvement in run time, whilst maintenance in accuracy, indicates the power of the surrogate model and its potential usage in clinical settings.

5.5 Conclusion

Once evaluating the results, there is a clear indication of the project's goal being reached. The models for the one and two chambers were implemented and sensitivity analysis was performed on it. The results indicate the run time was too long for large samples which is shown in table 5.3 and therefore a surrogate model was built. The surrogate sustains the accuracy whilst improving the computational time exponentially and thus the run time was improved and the project aims and objects were met, which were outlined in chapter 3.

Table 5.3: The Run Time of Models with the Number of Samples Used.

Model	**Sample Size**	**Computational Time (s)**
One Chamber	500	98.79
One Chamber	1500	285.34
One Chamber Surrogate	500	0.35
One Chamber Surrogate	1500	2.60
Two Chamber	460	311.15
Two Chamber	9200	10695.88
Two Chamber	23000	34636.03
Two Chamber Surrogate	460	2.27
Two Chamber Surrogate	9200	13.91
Two Chamber Surrogate	23000	112.66

Chapter 6

Conclusions and Future Work

6.1 Future Work

The project has achieved its aims and objectives as outlined in Chapter 1 and Chapter 3. For future possible work for further implementation it is possible to investigate the different techniques that were outlined beforehand for each sections from the sensitivity analysis method, to the surrogate model. The techniques could be applied to the four chamber model as well.

6.1.1 Sensitivity Analysis

It is possible to use different type of sensitivity analysis methods and compare their performance to each other. In the literature review in chapter 2, there is an outline of various methods. For possible future works, there is an opportunity to analyse the model with different techniques and compare which method brings about the most accurate and efficient analysis and perhaps review the trade off for using one technique over the other, for example using morris' method, sobol, local, efast.

6.1.2 Surrogate Model

The same can be said about surrogate models. It would be possible to implement different techniques to view the various outcomes. For example, if time constraint was not an issue, it would be ideal to implement Kriging, neural networks, bayesian regression etc. It would provide insight to the various runtimes and accuracy.

6.2 Conclusion

The project has achieved the ability to model the one chamber and two chambers of the heart, applied sensitivity analysis to the models and formed surrogate models of the heart. Thus, reducing its computational time. The scope of this project has been difficult as not only was it extremely hard to understand concepts not taught before, sensitivity analysis and surrogate modelling, but was also hard to implement the ideas in to code. The libraries used for coding were difficult to comprehend and there were many parameters to consider, the numerous libraries being used and the theory of the project itself. Understanding this, the reading for the project's literature began in the summer to get an understanding on the project. This project has been the most difficult project undertaken, but the journey to comprehension and the end result was very much satisfying.

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Appendices

Appendix A

Appendix

$$R = \frac{(P_{\text{ao,mean}} - P_{\text{ven,mean}})}{\text{CO}} \approx \frac{P_{\text{ao,mean}}}{\text{CO}} \tag{A.1}$$

This equation is the representation of resistance in the cardiovascular system. It uses the concept of Ohm's law. $P_{ao,mean}$ is the mean arterial pressure in the aorta, $P_{ven,mean}$ is the mean venous pressure, typically much smaller than $P_{ven,mean}$ and often approximated as negligible and CO is the cardiac out, representing the volume of blood the heart uses per unit of time.

$$C = \frac{\Delta V}{\Delta P} \tag{A.2}$$

The equation uses the compliance of the blood vessel. ΔV is the change in volume of the vessel and ΔP is the pressure difference between systolic and diastolic phases. C is simply the compliance of the vessel.

$$\frac{dP}{dt} = \frac{1}{C}\left(Q_{\text{in}} - \frac{P}{R}\right) \tag{A.3}$$

This is the differential equation for the 2-element Windkessel model describing the dynamics of pressure in a compliant system in terms of flow in, resistance and compliance. The equation balances pressure increase caused by inflow Q_in and pressure drops caused by outflow (P/R) and is regulated by compliance (C).

$$\frac{dP}{dt} = \frac{1}{C}\left(Q_{\text{in}} - Q_{\text{out}}\right) \tag{A.4}$$

where:

$$Q_{\text{out}} = \frac{P}{R + Z} \tag{A.5}$$

The dynamics of pressure P in a cardiovascular vessel in terms of inflow (Q_in) and outflow (Q_out) of blood. $\frac{1}{C}$ converts volume changes (from inflow and outflow) into pressure changes. Q_out is pressure inside the vessel (P) divided by resistance to blood flow (R) and impedance

(Z), which accounts for resistance to flow. This is for the 3-element Windkessel model.

$$L\frac{dQ_{\text{in}}}{dt} + Q_{\text{in}}R + \frac{P}{C} = 0 \tag{A.6}$$

This equation incorporates blood flow (Q_in) and pressure (P) in a cardiovascular system and includes the effects of inductance (L). $L\frac{dQ_{\text{in}}}{dt}$ is the inertial forces where it overviews the resist in changes in flow rate. $Q_{\text{in}}R$ is the resist of the flow due to friction. $\frac{P}{C}$ is the reflection of the restoring force of the vessel wall due to pressure. This is the balance of forces expressed in the 4-element Windkessel model.

$$Y = G(x) \tag{A.7}$$

This is the basis of sensitivity analysis, where Y is the model output, $G(x)$ is the model function and $x = [x_1, x_2, \ldots, x_n]$ is the vector of input parameters.

$$EE = \frac{f(\theta + \Delta) - f(\theta)}{\Delta}, \quad \Delta = \frac{l}{2(l-1)}, \tag{A.8}$$

This is the main aspect of Morris's Method of sensitivity analysis, calculating the elementary effect (EE) of a single input parameters on the model output, it quantifies the local sensitivity of the model's output f to a single input parameter change. $f(\theta)$ is the model output with θ as the model's input. $f(\theta + \Delta)$ is the model output with the change from (Δ) is applied to θ. Δ is the small change applied to the parameter.

$$x \in D_X \subset \mathbb{R}^d \mapsto y = G(x) \tag{A.9}$$

which is constructed based on a limited number of runs of the true model, the so-called experimental design:

$$\mathcal{X} = \{x^{(1)}, \ldots, x^{(n)}\}. \tag{A.10}$$

$x \in D_X$ is x as a vector of input parameters from a domain D_X which is a subset of $\mathbb{R}^d$. d is the dimension of the input space and D_X is the valid rang for the input parameters. This all maps to y which is the model output and $G(x)$ is the simulation function, mapping the input variables x to output y.

$\mathcal{X}$ is the experimental design, being the finite set of input variables $\{x^{(1)}, \ldots, x^{(n)}\}$ in the input domain D_X. x^i is a specific vector of input variables where the simulation function is evaluated.